

CHARLES UNIVERSITY
FACULTY OF SOCIAL SCIENCES
Institute of Economic Studies



**Explaining Czech Inflation with a
Bernanke-Blanchard Framework:
An Exploratory Price-Level Extension**

Master's thesis

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Declaration of Authorship

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Prague, May 4, 2026

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Abstract

This thesis analyzes the drivers of post-pandemic inflation in the Czech Republic using the semi-structural framework of Bernanke and Blanchard (2023). The primary goal of this analysis is to provide a country-specific application of the model, ensuring comparability with existing international evidence. As an exploratory extension, the framework is further augmented by incorporating a proxy for the domestic price level into the price equation.

The empirical results indicate that the inflation surge was primarily driven by external cost-push shocks with a relatively strong pass-through to domestic inflation. The persistence of inflation remains substantial, pointing to backward-looking price-setting behavior. Labor market tightness is found to be an important driver of wage dynamics; however, the estimated catch-up effect is negative, suggesting that past inflation surprises did not translate into higher nominal wage growth.

Regarding the exploratory extension, the specification including the price level gap does not provide statistically significant evidence of state-dependent amplification of external shocks. This result is robust across alternative specification.

These findings imply that inflation can remain high even in the absence of strong second-round wage effects.

JEL Classification E31, E52, E32, E37, E58

Keywords Inflation, Price shocks, Price level, Inflation Dynamics

Title Explaining Czech Inflation with a Bernanke-Blanchard Framework: An Exploratory Price-Level Extension

Abstrakt

Tato práce analyzuje determinanty post-pandemické inflace v České republice pomocí semi-strukturálního modelu Bernankeho a Blancharda (2023). Hlavním cílem analýzy je aplikace tohoto modelu na česká data, což umožňuje mezinárodní srovnatelnost výsledků. V rámci průzkumného rozšíření je rámec dále obohacen o proměnnou zachycující domácí cenovou hladinu v rovnici pro ceny.

Empirické výsledky naznačují, že inflační epizoda byla primárně tažena externími nákladovými šoky s relativně silným přenosem do domácí inflace. Inflace vykazuje vysokou míru setrvačnosti, což naznačuje převážně zpětně hledící cenotvorbu. Napjatost trhu práce se ukazuje jako důležitý faktor mzdové dynamiky, nicméně odhadnutý catch-up efekt je záporný, což naznačuje, že neočekávaná inflace se nepromítla do vyššího růstu nominálních mezd.

Pokud jde o průzkumné rozšíření specifikace zahrnující mezeru cenové hladiny, ta neposkytuje statisticky významné důkazy o multiplikačním efektu při přenosu externích šoků. Tento výsledek zůstává robustní i v alternativní specifikaci.

Tyto poznatky naznačují, že inflace může zůstat zvýšená i bez přítomnosti silných sekundárních mzdových efektů.

Klasifikace JEL	E31, E52, E32, E37, E58
Klíčová slova	Inflace, Cenové šoky, Cenová hladina, Dynamika inflace
Název práce	Vysvětlení české inflace pomocí Bernanke-Blanchardova modelu: Průzkumné rozšíření o cenovou hladinu

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Acronyms

CPI	Consumer Price Index
US	United States
ECB	European Central Bank
UK	United Kingdom
BB	Bernanke-Blanchard
EA	Euro area
EU	European Union
GDP	Gross Domestic Product
PC	Phillips Curve
G7	Group of Seven
OECD	Organisation for Economic Co-operation and Development
EMU	Economic and Monetary Union
NEER	Nominal Effective Exchange Rate
BS	Balassa-Samuelson
V4	Visegrad Four
CEE	Central and Eastern Europe
SVAR	Structural Vector Autoregression
IRF	Impulse Response Function
GAP	Price Level Gap
CZSO	Czech Statistical Office
CNB	Czech National Bank
LFS	Labour Force Survey

ESA	European System of Accounts
HICP	Harmonised Index of Consumer Prices
GSCPI	Global Supply Chain Pressure Index
MoLSA	Ministry of Labour and Social Affairs
HFCE	Household Final Consumption Expenditure
PLI	Price Level Index
NYFED	Federal Reserve Bank of New York
TVP	Time-Varying Parameter
SD	Standard Deviation
GNI	Gross National Income
OLS	Ordinary Least Squares
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
ADF	Augmented Dickey-Fuller
SOE	Small Open Economy

Master's Thesis Proposal

Author	Václav Šmíro
Supervisor	PhDr. Jiří Schwarz, Ph.D.
Proposed topic	Explaining Czech Inflation with a Bernanke-Blanchard Framework: An Exploratory Price-Level Extension

Motivation Recently, there has been a surge in inflation in advanced economies, driven by factors such as deferred consumption and supply chain disruptions, or the outbreak of Russian aggression in Ukraine. The latter caused energy cost shocks. Additionally, we observe higher inflation rates in Eastern European countries compared to their Western counterparts. This discrepancy can be attributed to the greater energy demands and stronger reliance on supply chains with Russia in the East.

We can see that the higher inflation resulting from cost shocks has spilled over into the core inflation of services. Notably, this core inflation tends to be higher in countries with lower relative price levels for services. One reason for this may be the price level itself, which is the motivation for this thesis, because if countries with lower price levels face the same cost shock as countries with higher price levels, the shock and the follow-up will be reflected in a higher percentage level. An example is the simple model described by Schwarz & Adam (2023) in their CNB article, where a \$1 increase in the cost of lunch may have a more substantial impact on inflation in a country where lunch costs \$5 (a 20% increase) than in a country where it costs \$10 (a 10% increase). This raises another interesting question: How much quantitatively does the starting price level affect the inflation rate? If it is an important factor, how much should the central bank respond to this inflation surge in Eastern Europe by adjusting interest rates? Is it appropriate to follow the same approach as Western European countries, considering that inflation does not necessarily imply economic overheating?

For this reason, I have decided to focus on exploring potential determinants of inflation in Czechia using model provided by Blanchard & Bernanke (2023). Further to put Czechia in context, the thesis will also focus on differences in inflation across the

European Union and analyze potential reason, why is inflation surge differs between countries.

The literature gives several reasons why inflation differentials between countries occur. One is the differential response to common shocks and asymmetric shocks. According to Bank (2003) the determinant may be the difference in the business cycle. This is also said by Andersson *et al.* (2009), where he found that the different position of the business cycle is the main driver. Another reason may be the different structure of the economy, where for example when energy prices change, countries in higher energy intensity face higher inflation (Beck *et al.* 2009). This occurs because energy wholesale market have been transmitted through production affecting non-energy items such as services or goods. Asymmetric shocks take place when the nominal exchange rate changes, even for countries that share the same currency, because some countries are more oriented to foreign markets than others and thus are exposed to the nominal exchange rate more (Beck *et al.* 2009). Another possible explanation according to Borio & Filardo (2007), is the degree of globalization. Countries with a higher degree of globalization have a common inflation driver and vice versa. For example, Ciccarelli & Mojon (2010) examined inflation variability between OECD advanced economies and discovered that almost 70% is driven by common factors, but in the recent large increase in inflation, the influence of domestic factors can be noted (Binici *et al.* 2022).

Furthermore, countries may differ according to the composition of consumption, and this has an impact on the amount of inflation if food and energy have a greater weight in the consumption basket. This is also related to the different specialization in sectors, where countries with higher production intensity are exposed to higher inflation. In contrast, countries with a higher share of GVA in services have lower inflation rates (Coutinho & Licchetta 2023).

Another reason for the divergence in inflation rates is the structural differences between markets and labor markets. In this regard, Beck *et al.* (2009) say that differences in the structure of labor markets and differences in structural unemployment lead to inflation differentials. At the same time, different levels of competition in the labor market and local structural factories can also lead to different inflation rates (Bank 2003). It should not be overlooked that the different energy market structure had an impact on the inflation rate in 2022. Countries which regulated energy price during the high energy price had lower pass-through, additionally, countries have different percentages of contracts that help to slow down the pass-through (Corsello & Tagliabracci 2023). Also, local differences in margins can lead to inflation differentials. For example, in 2022, when energy prices were surging, higher margins were also recorded in this sector. However, it is difficult to know at the aggregate level whether there was an increase in margins or marginal costs (Corsello & Tagliabracci

2023; Hernnäs *et al.* 2023).

There is further inflation differentials if people have different inflation expectations since this plays an important role in pricing. This unanchoring can lead to second round effect on wages (Boissay *et al.* 2022). Thus, menu cost has less and less effect and higher prices become acceptable (Buelens 2023). Also, it is necessary to add that in times of high inflation, people follow more past inflation than expected inflation, due to higher uncertainty (Baba *et al.* 2023).

The reasons for inflation differentials are also institutional factors such as the degree of regulation (Andersson *et al.* 2009), centralization or decentralization, where countries with strong centralization or decentralization are more capable of dealing with cost shocks than countries with a moderate degree of centralization (Calmfors & Driffill 1988). If the market is at perfect competition, inflation differentials should converge to zero (Gautier *et al.* 2022).

The last of a variety of reasons for the divergence in inflation rates may be nominal convergence, where relatively poor countries experience higher inflation than rich ones. The explanation for higher inflation may be the appreciation of the real exchange rate as labour productivity rises for tradable goods, then for non-tradable goods, resulting in higher prices of non-tradable goods relative to tradable goods as wages are equalized across sectors (Balassa 1964; Samuelson 1964). As you can see, there are many possible reasons and I would try to quantify the strength primarily on the starting price level.

Hypotheses

Hypothesis #1: There is no significant difference in the response of core inflation to commodity price shocks between Eastern and Western European countries. It is conditional on the type of shock?

Hypothesis #2: Prices in the Czech Republic are persistent to shocks (price stickiness)

Hypothesis #3: Labor tightness was a significant factor contributing to higher Czech inflation

Methodology The main pillar of the thesis will be the application of the macroeconomic model introduced by Bernanke & Blanchard, which aims to explain the cause of the recent high inflation and to find out what transmission channels have the biggest impact. I apply the model to Czech data, using four equations for wages, inflation and inflation expectations, which are split into long-run and short-run inflation expectations. For wages and price inflation, I use the Average Gross Monthly Wage (quarterly frequency, Nominal Wage Index y-o-y=100) and the Consumer Price

Index according to ECOICOP (Monthly, 2015, Average of base year = 100), published by the Czech Statistical Office. Inflation expectations, which are divided into one-year and three-year horizons, are based on quarterly data considering the financial market, constructed by the Czech National Bank.

In addition, we have exogenous variables, that are used to try to explain the variables mentioned above (more detailed equation structure will be presented in the outline). As cost shocks, is considered the growth in energy and food prices. These data are a subcomponent of the consumer price index published by the Czech Statistical Office. Furthermore, among other cost shocks are shortages of goods due to changes or breaks in supply chains. These data are obtained from the Global Supply Chain Pressure Index (GSCPI) published by the FED. To determine the effect of labor tightness on the level of inflation I use the number of unplaced job seekers divided by number of job vacancies published by The Ministry of Labor and Social Affairs.

The thesis will also focus on differences in inflation rates across the European Union and analyze potential reason, why is inflation surge different between countries such as geographical location, market monopolization and others. One of the possible causes of disparity could be the price level. In order to assess the impact of the price level, a decomposition of cost shocks will be performed and then compared with other central bank results that have used the same model. In order to make comparisons across countries easier, the model is modified to use the Harmonised Index of Consumer Prices (HICP) instead of the CPI. Both price indices are therefore included in the model for better usability for the Czech National Bank.

Expected Contribution The thesis applies the model presented by Bernanke & Blanchard to Czech data and helps to explain the relationship between price shocks, wage inflation, long-run and short-run inflation expectations in the Czech environment, these results will help to understand the transmission mechanism of inflation and help the Czech National Bank to better respond to price shocks and maintain its price stability mandate. We can also see that although some papers mention the effect of the price level on inflation, in my opinion it is rather omitted and should be given more weight in central bank decision making.

Outline

1. Inflation overview around the world (2019-2024)
2. Literature overview on inflation differentials
3. What could cause inflation surge?

- Strong aggregate demand (pent-up savings)

- Expansionary fiscal policy
- Accommodative monetary policy
- Sharp increase in commodity prices
- Supply chain disruptions
- Reduction in labor supply

4. **Transmission mechanism of inflation**

- Inflation due to supply factors
- Inflation due to strong demand would lead to sustained upward pressure on inflation
- Current focus in on to get inflation back to central bank targets “ will it require cooling of the labor market?

5. **Model introduced by Bernanke & Blanchard**

- Include 4 equations, corresponding to 4 endogenous variables
 - Price inflation
 - Wage inflation
 - SR inflation expectations
 - LR inflation expectations
- Model is divided on analytical model and empirical counterpart
- **The wage equation**
 - Wage inflation depends on lagged wage inflation, lagged expected short-run inflation, the lagged ratio of vacancies to unemployment, the lagged catch-up variable, a productivity growth rate and two dummies to capture the unusual movements in wages in 2020Q2 and 2020Q3
- **The price equation**
 - Price inflation depends on lagged price inflation, lagged wage inflation, a productivity growth rate, and three measures of price shocks: the rate of growth of the price of energy relative to the wage, the rate of growth of the price of food relative to the wage, and a measure of shortages, based on Google trends
- **The SR inflation expectation equation**
 - Short-run inflation expectations depend on themselves lagged, long-run expectations, and on current and lagged actual inflation. There is no constant term.

- **The LR inflation expectation equation**

- Long-run inflation expectations depend on themselves lagged, and on current and lagged actual inflation. There is no constant term.
- The four equations are constrained by homogeneity restrictions, which imply no long-run tradeoff between inflation and activity
- Identification was achieved through restrictions on contemporaneous coefficients, notably the assumption that wage inflation does not react to contemporaneous movements in its determinants. This approach is similar to a structural VAR.

Impulse response function of analytical model:

- **Dynamic effect of energy price shock on inflation**

- The direct impact on consumer prices
- The effect on nonenergy goods and services
- Influence on inflation expectations
- The subsequent increase in production costs

- **Dynamic effect of a permanent increase in labor market tightness on inflation**

- The direct effect on wage inflation
- The speed of wage pass-through to prices
- The influence on inflation expectations
- the subsequent impact on wage inflation and price inflation
- this effect is more persistent
- While the contribution of tight labor conditions to inflation is small in the short run, it becomes more significant as the price shocks dissipate over time.
- **Inflation decomposition in Czechia**
 - To understand the sources of inflation, a decomposition of inflation into its components of origin, such as increases in food prices, energy, shortages, productivity and the labor market, is carried out.

6. Inflation in EU countries

- Differences in the inflation surge between countries
- Elaborate the possible reasons for the differences in inflation surge
 - high energy intensity

- lower GDP
- monopolization
- **price level**

7. Does the price level matter?

- Effect of absolute price shocks in economy with lower price level
- Simple regression model between HCPI and price level

8. Conclusion

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Chapter 1

Introduction

For more than two decades, advanced economies operated in an environment of remarkable price stability, described as the Great Moderation. The post-pandemic recovery, however, ended this era, plunging the global economy into the most severe inflationary episode in recent decades. This surge in inflation not only challenged the credibility of central banks, but also caused a cost-of-living crisis. The consequences were severe for households. According to Pallotti *et al.* (2023), the welfare loss, measured by the decrease in real disposable income, reached 3% in France and Spain, 7% in Germany and 9% in Italy.

Although the initial supply chain disruptions and the subsequent Russian invasion of Ukraine were global, the macroeconomic consequences were highly asymmetric. For example, while 12-month inflation in Euro area (EA) remained below 5.0% until December 2021, the United States (US) had already crossed that threshold six months earlier. Even more striking was the heterogeneity within Europe itself. At the peak of the crisis in October 2022, inflation in France reached a relatively contained 6.6%, while converging economies such as Estonia experienced staggering rates of over 25.2% (Bernanke & Blanchard 2024). Governments scrambled to mitigate these effects; without unconventional fiscal subsidies and energy price caps, the peak 12-month inflation in EA could have been almost 3 percentage points higher (Dao *et al.* 2023).

This extreme variance across countries highlights the need to understand inflation dynamics at the national level, particularly in converging economies like the Czech Republic, which experienced one of the highest inflation rates in the EU. To systematically uncover the drivers of this episode, the primary objective of this thesis is to apply the semi-structural framework developed

by Blanchard & Bernanke (2023) to Czech data. By estimating the model on quarterly data from the end of 2000 to 2025, we aim to provide a detailed historical decomposition of Czech inflation, isolating the roles of external price shocks, labor market tightness, and inflation expectations.

As a second, exploratory step, this thesis extends the standard Bernanke-Blanchard (BB) framework to investigate whether the specific characteristics of a converging economy play an amplifying role. Specifically, we explore the hypothesis that a structurally lower domestic price level may amplify the pass-through of external commodity shocks, as globally priced commodities constitute a larger share of marginal costs. While a formal cross-country identification of this mechanism would require a panel design, we utilize the Czech Republic as a case study to examine whether such state-dependent amplification is present in a single-country setting.

Our findings provide evidence on the domestic drivers of Czech inflation. The results suggest that while energy price shocks were the main trigger – with an estimated pass-through coefficient in the Czech Republic (0.09) more than double that of EA (0.04) – the persistence of inflation was sustained by domestic imbalances. Furthermore, the estimates indicate significant price stickiness (0.74), suggesting that, while workers are forward-looking, Czech firms remain backward-looking in their price-setting. The historical decomposition also clarifies the specific role of the labor market: extreme pre-pandemic tightness created a persistently high underlying baseline for inflation, even though post-pandemic shifts in the V/U ratio acted as a relative cooling factor against massive commodity spikes. Finally, regarding the exploratory extension, the inclusion of the price-level gap does not yield statistically significant evidence of state-dependent amplification.

The remainder of this thesis is structured as follows. Chapter 2 reviews the existing literature and replications of the BB model. Chapter 3 introduces the theoretical framework and formalizes the extension of the price level. Chapter 4 describes the data set and the variables. Chapter 5 details the empirical methodology. Chapter 6 presents the estimation results, impulse response functions, and historical decompositions. Chapter 7 discusses the policy implications for central banks in converging economies. Finally, Chapter 8 concludes the thesis.

Chapter 2

Literature Review

2.1 General Drivers of Post-Pandemic Inflation

The global economy experienced an unprecedented surge in inflation starting in 2021, marking a sharp departure from the low-inflation environment of the previous two decades. The literature generally agrees that this inflationary episode was driven by a complex interplay of negative supply shocks and demand-side stimulation (Baba *et al.* 2023). Initially, pandemic-induced lockdowns severely disrupted global supply chains, creating severe bottlenecks in the manufacturing and shipping sectors. These constraints coincided with a strong, policy-driven recovery in aggregate demand, fueled by expansionary monetary and fiscal policies in advanced economies.

As the global economy began to recover, a secondary and more severe shock emerged in early 2022 with the Russian invasion of Ukraine. This geopolitical crisis caused massive spikes in global commodity prices, particularly in the energy and food markets. According to recent consensus, the combination of these consecutive shocks created a unique constellation of adverse factors: supply chain disruptions initiated price increases, while energy shocks further amplified them and incorporated inflationary pressures into broader Consumer Price Indices (CPIs). Properly identifying the relative contributions of supply versus demand factors, as well as the specific role of the labor market in sustaining these price increases, nonetheless remains a complex task. To address this challenge and systematically trace the transmission of these impulses into domestic prices and wages, this thesis adopts the semi-structural framework of Blanchard & Bernanke (2023).

2.2 International Evidence and Replications of the BB Model

As mentioned previously, Blanchard & Bernanke (2023) introduced a dynamic semi-structural framework specifically designed to decompose the drivers of the post-pandemic inflation surge in the US. Their analysis concluded that while the initial inflation surge in 2021 was primarily driven by commodity price shocks and sectoral supply-demand imbalances, the persistence of inflation was increasingly sustained by labor market tightness and rising nominal wage growth. This methodology has since become a benchmark, leading to a coordinated research effort across numerous central banks to identify domestic drivers of inflation.

Replications of the model in the EA reveal significant differences from the experience in the US, particularly with respect to the role of the labor market. For Germany and the broader EA, Menz (2024) finds that inflation was almost exclusively a supply-side phenomenon driven by energy costs, with labor market tightness playing a substantially smaller role than in the US. Similarly, Pisani & Tagliabracchi (2023) demonstrate that for the Italian economy, energy price shocks (both direct and indirect) were the dominant force, while the wage-price spiral remained muted due to anchored expectations and specific institutional settings.

In France, Aldama *et al.* (2024) argue that while energy and food shocks were the primary triggers, the effective anchoring of inflation expectations by the European Central Bank (ECB) prevented these shocks from translating into a self-sustaining wage-price spiral. This contrast is further highlighted by the case of Belgium. De Walque & Lejeune (2024) show that Belgium's unique system of automatic wage indexation led to higher inflation persistence compared to its neighbors, making the economy more vulnerable to the initial commodity price spikes. In Spain, Ghomi *et al.* (2024) emphasize the role of fiscal policy and government interventions, such as energy price caps, which successfully decoupled parts of the domestic price index from international market volatility.

Beyond the EA, the results show further heterogeneity. Haskel *et al.* (2023) find that United Kingdom (UK) inflation was particularly "sticky" compared to the US, with food price shocks having an especially prolonged impact. For Canada, Bounajm *et al.* (2024) find results that most closely mirror the US

experience. In the Netherlands, Bonam *et al.* (2024) point to the combined pressure of commodity shocks and an historically tight labor market as the primary drivers of Dutch price dynamics. Perhaps the most distinct application is provided by Nakamura *et al.* (2024), who use the framework to analyze Japan's exit from a decades-long deflationary environment.

While these studies provide a comprehensive overview of advanced economies, there is a notable gap regarding the application of this framework to converging economies in Central and Eastern Europe. To fill this gap, our thesis applies the BB framework to the Czech Republic, aiming to determine whether its inflation dynamics align more closely with the supply-driven European experience or the demand-driven US model.

2.3 Determinants of Inflation Differentials in Open Economies

While the BB framework provides a structural method for decomposing inflation within a single country, it does not fully explain why identical global shocks result in widely varying inflation rates across different economies. To understand these disparities, particularly in the context of the European Union (EU) and converging economies like the Czech Republic, it is necessary to examine the underlying drivers of inflation differentials. Reviewing these structural factors is crucial for our thesis, as they motivate the rationale for our exploratory extension concerning the domestic price level.

2.3.1 Response to Common Shocks and Economic Structure

According to Andersson *et al.* (2009), one of the main factors influencing the differences in inflation between countries is the variation in the business cycle position, often measured by the output gap. Countries with a positive output gap tend to have higher inflation due to increased labor demand and rising wages, a relationship classically described by the Phillips Curve (PC). Furthermore, the effect of the business cycle on inflation differentials is confirmed by Rogers (2001) and Liargovas & Arvanitis (2023), who note that a significant part of the variation can be attributed to the cyclical position of the country relative to others. This cyclical variation justifies the inclusion of labor market tightness (V/U ratio) as a core variable in our empirical baseline model.

Beyond cyclical positions, inflation differentials often result from the different structural characteristics of economies. A key factor is energy intensity. In 2022, for example, increases in energy prices explained a massive share of the inflation surge in the EA, with industrial-oriented countries being significantly more affected than service-oriented ones (Coutinho & Licchetta 2023; Bigerna 2023). Additionally, varying degrees of market competition and nominal rigidities complicate the adjustment of economies to external shocks (Stylianou 2023). For a highly industrialized economy like the Czech Republic, capturing these direct and indirect commodity price pass-throughs is essential, which is why energy and food price shocks are explicitly modeled in our specification.

2.3.2 Globalisation and Inflation Expectations

Globalization has fundamentally altered inflation dynamics over the last decades. Borio & Filardo (2007) conclude that global factors play an increasingly important role, often overshadowing domestic measures of economic slack in traditional PCs. Istiak *et al.* (2021) show that inflation spreads between countries through trade channels, while Ciccarelli & Mojon (2010) view inflation largely as a global factor. Conversely, Binici *et al.* (2022) suggest that during the post-pandemic period, the role of domestic factors has resurged as consumer behavior and local constraints became dominant again. Our approach allows us to separate these global supply shocks from domestic demand pressures.

Equally important is the role of inflation expectations. Buelens (2023) points out that in periods of low inflation, agents pay little attention to price developments. Once inflation crosses a certain threshold, their attention increases, potentially anchoring higher expectations and changing pricing behavior. To account for this potential regime shift and its role in sustaining inflation, our model incorporates explicit measures of both short-term and long-term inflation expectations into the wage and price equations.

2.3.3 Nominal Convergence and the Price Level

A traditional explanation for structural inflation differentials in catching-up economies is nominal convergence. Historically, the Balassa-Samuelson (BS) effect suggests that productivity catch-up in the tradable sector leads to higher overall inflation as wages equalize across sectors (Balassa 1964; Samuelson 1964).

Empirical evidence on the BS effect in Europe is mixed. While De Haan (2010) argues that nominal convergence is a primary driver, studies like Honohan *et al.* (2003) and Beck *et al.* (2009) question its practical significance, finding no strong relationship between initial income levels and subsequent price changes. Similarly, Egert (2007) show that the BS effect explained at most one-third of the real exchange rate appreciation in the Visegrad Four (V4) countries between 1995 and 2006. Coutinho & Licchetta (2023) analyze the 2022 energy crisis and identify the BS effect as a limited long-run factor, highlighting short-term drivers like energy dependence as more critical.

Building on this literature, our thesis approaches nominal convergence from a different angle. Rather than focusing on long-term productivity differentials (the classical BS effect), we explore whether the structurally lower price level in a converging economy acts as an amplifier during periods of severe global shocks.

2.3.4 Summary and Motivation for the Extension

Recent macroeconomic debates have increasingly focused on non-linearities in the PC. While most studies, such as Benigno & Eggertsson (2023), attribute this non-linearity to capacity constraints in the labor market, converging economies may experience a different type of non-linearity: a state-dependent pass-through of external shocks. In this context, a structurally low domestic price level can potentially act as an amplifier. When a global nominal shock occurs (e.g., a surge in energy prices), firms in economies with lower relative price levels might use the general inflationary environment as a catalyst to rapidly adjust their prices towards the international core level. This theoretical possibility motivates the exploratory extension of this thesis, where we introduce a price-level gap proxy into the standard BB framework to test for state-dependent amplification in the Czech data.

Chapter 3

Theoretical Framework

This chapter establishes the theoretical foundation for analyzing the transmission of inflationary shocks, moving from general propagation mechanisms to the specific semi-structural framework used in this study.

3.1 Transmission mechanism

The transmission mechanism of inflation describes the dynamic process through which initial economic shocks propagate through the economy. In the context of the recent post-pandemic inflationary episode, it is crucial to distinguish between the direct, immediate impacts of exogenous shocks and their indirect, second-round effects.

Methodologically, analyzing this transmission requires a framework capable of capturing the loops between different macroeconomic variables. For this reason, one of the empirical approaches to inflation dynamics are closely related to Structural Vector Autoregression (SVAR) models. As noted by Gottschalk (2001), the SVAR methodology is specifically designed to transform reduced-form residuals, which are often mutually correlated and lack a direct economic interpretation into structural shocks that are mutually orthogonal and represent distinct economic events, such as supply or demand shocks.

Similar to an SVAR approach, the transmission mechanism in this study is analyzed by estimating a system of dynamic, interconnected equations. The identification of this system requires imposing structural restrictions based on economic theory to isolate the contemporaneous relationships between variables. Once identified, the model utilizes Impulse Response Functions (IRFs) to trace the path of a specific, isolated shock over time. These functions provide

the tool for interpreting the dynamic propagation of shocks, showing how a temporary disturbance in one variable (e.g., energy prices) gradually affects the entire system through its internal feedback structure (Gottschalk 2001).

3.2 The Bernanke-Blanchard Model

To formalize the transmission mechanism described above, we utilize the semi-structural framework developed by Blanchard & Bernanke (2023). The framework is analytically built upon three main pillars: wage-setting behavior, price-setting behavior, and the formation of inflation expectations.

3.2.1 Wage-Setting Behavior

Following the derivation in the original Blanchard & Bernanke (2023) framework, wage determination is modeled as a bargaining process between workers and firms. Workers negotiate to achieve a target nominal wage (w_t^*) that maintains their desired real purchasing power and reflects their labor productivity trend ($z_{w,t}$). They base their demands on expected future inflation (p_t^e). Furthermore, if past actual inflation (p_{t-1}) exceeded their past expectations (p_{t-1}^e), workers demand an additional "catch-up" wage increase to compensate for the unexpected loss in real income.

The target nominal wage can thus be formulated as:

$$w_t^* = p_t^e + \alpha(p_{t-1} - p_{t-1}^e) + z_{w,t} \quad (3.1)$$

Where $\alpha \geq 0$ measures the degree to which workers are compensated for past inflation surprises.

The actual negotiated wage (w_t) deviates from this target depending on the relative bargaining power of workers. This bargaining power is represented by a function $f(x_t)$ that captures the state of labor market tightness x_t (typically measured by the ratio of vacancies to unemployment, V/U). As the labor market becomes tighter, workers can push the actual wage above their baseline target ($f'(x_t) > 0$). Substituting the target wage into the actual wage equation yields the baseline theoretical wage equation:

$$w_t = p_t^e + \alpha(p_{t-1} - p_{t-1}^e) + z_{w,t} + f(x_t) \quad (3.2)$$

3.2.2 Price-Setting Behavior

The price equation is derived from the standard assumption of monopolistic competition, where firms set their prices (p_t) as a desired mark-up (μ) over their nominal marginal costs (mc_t):

$$p_t = \mu + mc_t \quad (3.3)$$

In this stylized framework, an economy's marginal costs are primarily determined by two components. The first component is unit labor costs, which is the difference between nominal wage growth and labor productivity growth ($w_t - z_{w,t}$). The second component is the cost of intermediate inputs, particularly commodities and materials subject to global supply shocks, denoted as $z_{p,t}$.

Assuming a constant steady-state mark-up (which can be normalized to zero when modeling variables in growth rates), substituting these cost components into the mark-up equation yields the theoretical price-setting equation:

$$p_t = (w_t - z_{w,t}) + z_{p,t} \quad (3.4)$$

In the absence of external supply shocks ($z_{p,t} = 0$), price inflation perfectly tracks the growth in unit labor costs, implying that the labor share of income (and the real wage mark-up) remains constant in the long run.

3.2.3 Inflation Expectations Formation

The interaction between wages and prices critically depends on how inflation expectations are formed. The model distinguishes between short-term and long-term expectations to capture the concept of "anchoring."

Short-term inflation expectations (p_t^e) are modeled as a weighted average of long-term expectations (p_t^*) and recent actual inflation (p_{t-1}):

$$p_t^e = \delta p_t^* + (1 - \delta)p_{t-1} \quad (3.5)$$

The parameter $\delta \in [0, 1]$ measures the degree to which short-term expectations are anchored to the long-term outlook. If δ is close to 1, short-term expectations look through temporary price shocks. If δ is close to 0, expectations are highly adaptive and respond strongly to past inflation, thereby increasing the risk of a wage-price spiral.

Long-term expectations (p_t^*), in turn, depend on the central bank's official inflation target ($\bar{p}$) and past actual inflation:

$$p_t^* = \lambda \bar{p} + (1 - \lambda)p_{t-1} \quad (3.6)$$

Here, $\lambda \in [0, 1]$ reflects the credibility of the central bank. A highly credible central bank will keep long-term expectations firmly anchored at the target ($\lambda \approx 1$), preventing temporary cost-push shocks from permanently shifting the inflation trajectory.

3.3 Exploratory Price Level Extension

To address whether the aggregate price level matters for inflation transmission, we extend the baseline theoretical framework. As an exploratory hypothesis, we test whether a lower relative price level may amplify an economy's vulnerability to external supply shocks, as globally priced commodities form a larger share of domestic costs.

Capturing this mechanism requires a state-dependent model. Building on recent literature that demonstrates a non-linear PC driven by the economy's *cyclical* state (Benigno & Eggertsson 2023), we apply this concept of state-dependency to the economy's *structural* state. Specifically, we introduce the domestic price level gap as a non-linear amplifier for global supply shocks. While the extension is motivated by a structural interpretation, the empirical implementation should be understood as an exploratory, reduced-form approximation of this mechanism.

As derived in the previous section, the original Blanchard & Bernanke (2023) theoretical price equation asserts that price inflation (p_t) is driven by unit labor costs and exogenous supply shocks $z_{p,t}$:

$$p_t = (w_t - z_{w,t}) + z_{p,t} \quad (3.7)$$

The microeconomic foundation of our extension lies in the endogenization of cost weights. In the standard Blanchard & Bernanke (2023) model, the pass-through of supply shocks is a constant parameter. We relax this by positing that the real cost share of globally-priced commodities is associated with the domestic price level. In economies with a large price level gap (GAP_t), locally denominated inputs (labor) are structurally cheaper, while commodities

$(z_{p,t})$ are priced at international levels. Consequently, commodities constitute a larger share of marginal costs. We formalize this by allowing the effective impact of a shock to vary: $(1 + \gamma GAP_t)z_{p,t}$.

By incorporating variable cost structures and structural convergence, we augment the theoretical price equation:

$$p_t = (w_t - z_{w,t}) + z_{p,t} + \gamma(z_{p,t} \times GAP_t) + \theta GAP_t \quad (3.8)$$

It is important to note that the interaction term should not be interpreted as isolating a single causal channel, but rather as capturing an association with differences in the economic environment correlated with the price level.

This specification aims to capture two distinct non-linear effects:

1. The Multiplier Effect of Commodity Shocks (γ): The interaction term $\gamma(z_{p,t} \times GAP_t)$ formalizes the hypothesis that the pass-through of global commodity shocks depends on the relative domestic price level. A larger gap acts as an amplifier ($\gamma > 0$) for external shocks, steepening the slope of the cost-push PC. As the economy converges ($GAP_t \rightarrow 0$), this amplification effect diminishes, and the sensitivity to shocks returns to the baseline level.

2. The Convergence Effect (θ): The standalone term θGAP_t introduces a direct structural component to domestic inflation. Economically, this is consistent with general price-level convergence. A country with a price level significantly below the core average will exhibit positive structural inflation ($\theta > 0$) as its productivity and non-tradable prices catch up over time. Including this term is also econometrically necessary to isolate the true multiplier effect of shocks from underlying price-level convergence trends.

It is important to note that the Price Level Gap (GAP) variable does not represent a purely structural measure of the price-level mechanism. Instead, it captures a combination of underlying structural characteristics typical of catching-up economies.

First, lower-price-level economies often exhibit a higher share of energy-intensive manufacturing in their Gross Domestic Product (GDP) compared to more service-oriented economies. Second, because domestic inputs such as labor are relatively cheaper, globally priced commodities constitute a larger share of marginal costs.

As a result, the estimated multiplier effect should be interpreted as a reduced-form relationship, reflecting a combination of price-level differences

and correlated structural factors, rather than a single identified causal mechanism.

Chapter 4

Data

This section describes the dataset used for the empirical model for the Czech Republic. The data are primarily sourced from the Czech Statistical Office (CZSO), the Czech National Bank (CNB), and the ECB. A comprehensive overview comparing the specific endogenous and exogenous variables used in this thesis with the original US model is provided in Appendix A.

The model is estimated using quarterly data covering the period from 2000Q4 to 2025Q3. Relative to the original BB (2023) study applied to the US, our estimation sample is shorter. This constraint is due to limited data availability, particularly regarding inflation expectations.

4.1 Endogenous variables

The set of endogenous variables in the model consists of five variables that capture the interaction between prices, wages, and expectations in the Czech Republic. These are summarized in Table A.1, and the nature of these data series is described in detail in the following subsections. Additionally, the catch-up term ($diff_cpi$), which measures the gap between the past year's inflation and previous expectations, is a constructed variable defined in detail in Chapter 5.

Table 4.1: List of Endogenous Variables and Data Sources

Variable	Description	Source	Transformation
<i>gw</i>	Wage growth (Compensation per employee)	ECB	Annualized QoQ growth
<i>gcpi</i>	Price growth (CPI)	CZSO	Seasonally adjusted, annualized QoQ growth
<i>cf1</i>	Short-term expectations (1-year ahead, non-financial firms)	CNB	—
<i>cf3</i>	Long-term expectations (3-year ahead, financial markets)	CNB	—
<i>diff_cpi</i>	Catch-up term (realized inflation surprise) as gap between past <i>gcpi</i> and past <i>cf1</i>	—	—

The historical development of the key endogenous variables over the estimation sample is depicted in Figure 4.1.

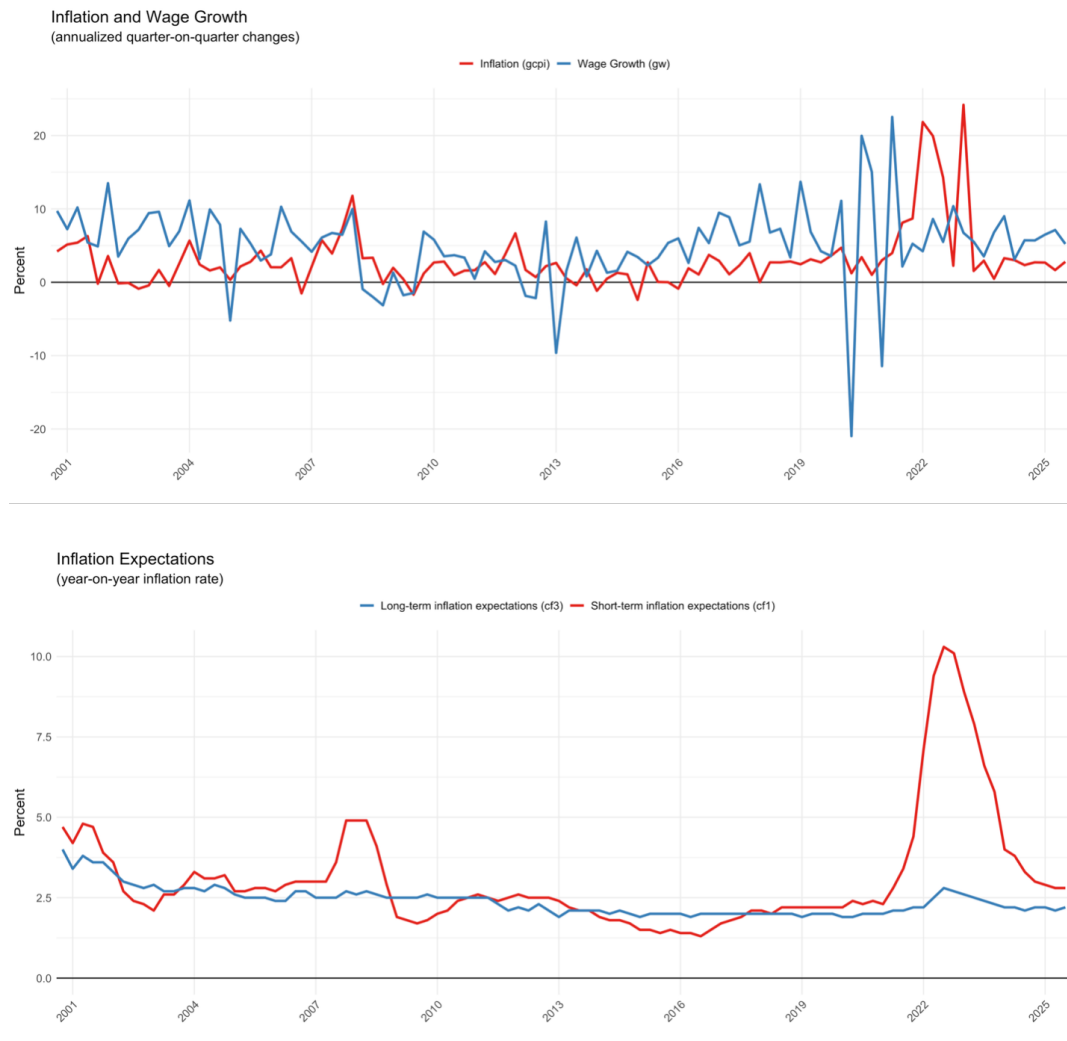


Figure 4.1: Development of Endogenous Variables

As depicted in Figure 4.1, both price and wage inflation remained relatively stable and subdued in the years preceding the pandemic. Nevertheless, starting in 2021, price inflation experienced an unprecedented surge. Wage inflation followed this upward trend but with a noticeable lag and at a lower magnitude. Regarding inflation expectations, the Czech data reveal a pattern consistent with the findings for the US (Blanchard & Bernanke 2023) and other European economies (Menz 2024). Short-term inflation expectations of non-financial firms rose sharply, closely tracking the trajectory of realized inflation. In stark contrast, long-term expectations of financial analysts remained remarkably well-anchored near the central bank's 2% target, exhibiting only a marginal and temporary elevation during the peak of the inflationary episode.

4.1.1 Wage Inflation

We measure wage growth using compensation per employee, which is defined as the total remuneration, in cash or in kind, payable by employers to employees in return for work (i.e., gross wages and salaries, bonuses, overtime payments, and employers social security contributions), divided by the total number of employees.

Alternatively, wage growth can be measured using compensation per hour worked. This measure was preferred by several European national banks when replicating the BB model (Menz 2024; De Walque & Lejeune 2024), as compensation per employee can be distorted by job-retention schemes (which were heavily utilized during the pandemic to stabilize employment) and is generally sensitive to employment composition effects. In the Czech environment, this composition effect is not a major issue, as the share of part-time employment is structurally very low compared to the EU average (Grossmann & Munich 2025).

Furthermore, when looking at the evolution of both variables in the Czech data (see Figure C.1 in Appendix), compensation per hour worked exhibits significant volatility compared to compensation per employee, particularly between 2014 and 2017. As noted in the official Inflation Reports (Czech National Bank 2014; 2015), this volatility was driven by the economic recovery, during which firms primarily absorbed new orders through the intensive margin, by abruptly increasing overtime and hours worked rather than hiring new employees. This introduced a high degree of fluctuation into the denominator of the hourly wage metric. Moreover, the transition to the European System of Accounts (ESA) 2010 methodology introduced substantial statistical noise, as data on hours worked from the Labour Force Survey (LFS) frequently diverged from business statistics (Czech National Bank 2014).

The estimation results for the wage equation using the compensation per hour worked are presented in Appendix (Table C.5) as a robustness check. Given the poorer fit of the model caused by the data anomalies mentioned above, we utilize compensation per employee as our primary measure in the baseline model.

4.1.2 Price inflation

Regarding price inflation, we utilize the quarterly annualized rate of change in the national CPI, seasonally adjusted and obtained from the CZSO¹. Our choice is motivated by several factors specific to the Czech institutional environment.

First, the national CPI is the official gauge for the CNB's inflation target (2%), making it the most relevant measure for analyzing monetary policy reactions and the formation of inflation expectations. Second, unlike the Harmonised Index of Consumer Prices (HICP), the Czech national CPI includes imputed rentals (owner-occupied housing costs), which represent a significant portion of Czech household expenditures. (Solc & Tomanova 2021)

We acknowledge that the price series experienced temporary statistical distortions in late 2022 due to the government's 'Energy Savings Tariff', which caused a significant artificial drop in the index, followed by a base-effect spike a year later (Czech National Bank 2022; Solc *et al.* 2023). Regarding the choice of index, one might ask why core inflation is not used. To maintain consistency with the methodology of the Bank of England (Haskel *et al.* 2023) and the Bundesbank (Menz 2024), we utilize the headline index, as headline prices are the primary reference for wage indexation and household expectations in the Czech Republic. For robustness, we also construct a wage equation using HICP data (see Appendix Table C.8).

4.1.3 Inflation Expectations

Short-term Inflation Expectations

For short-term inflation expectations, we use 1-year ahead data from managers of non-financial corporations, sourced from the CNB survey. These data are preferred as they better reflect the real economy's price-setting behavior. This is in contrast to consumer expectations, which tend to systematically overestimate inflation and exhibit a negative correlation between the economic outlook and price expectations (European Central Bank 2022).

¹The CPI series is constructed by merging two datasets due to the 2018 transition to the ECOICOP classification system. While the item breakdown changed, the index calculation methodology remained consistent, allowing for a continuous time series (Czech Statistical Office n.d.)

Long-term Inflation Expectations

Long-term inflation expectations are represented by the 3-year ahead expectations of financial market analysts. The primary reason for using financial analysts' forecasts is the limited data availability for the non-financial sector at a long horizon. Methodologically, these expectations are more forward-looking, well anchored to the inflation target, and serve as an indicator of the central bank's credibility. (Czech National Bank 1999)

4.2 Exogenous variables

The set of exogenous variables in the model consists of five series that capture external supply shocks and specific howlabor market conditions in the Czech Republic. These variables are summarized in Table A.2, and a detailed description of their construction and data sources is provided in the following subsections. This part of the model accounts for relative price shocks in energy and food, labor market tightness, global supply chain pressures, and productivity trends.

Table 4.2: List of Exogenous Variables and Data Sources

Variable	Description	Source	Transformation
$grpe_cpi$	Relative energy price growth	CZSO	Annualized QoQ growth
$grpf_cpi$	Relative food price growth	CZSO	Annualized QoQ growth
vu	Labor market tightness (ratio of job vacancies to unemployment)	CNB	—
$shortage$	Global Supply Chain Pressure Index	NYFED	Standardized
$gpty$	Labor productivity trend (per person)	ECB	Annualized QoQ growth, 8-quarter moving average
GAP_t	Price Gap	Eurostat	$100 - PLI_t$ (interpolation)
$GAPMA_t$	Trend Price Gap	—	8-quarter moving average of GAP_t

The historical evolution of these external supply shocks and specific labor market conditions is plotted in Figure 4.2.

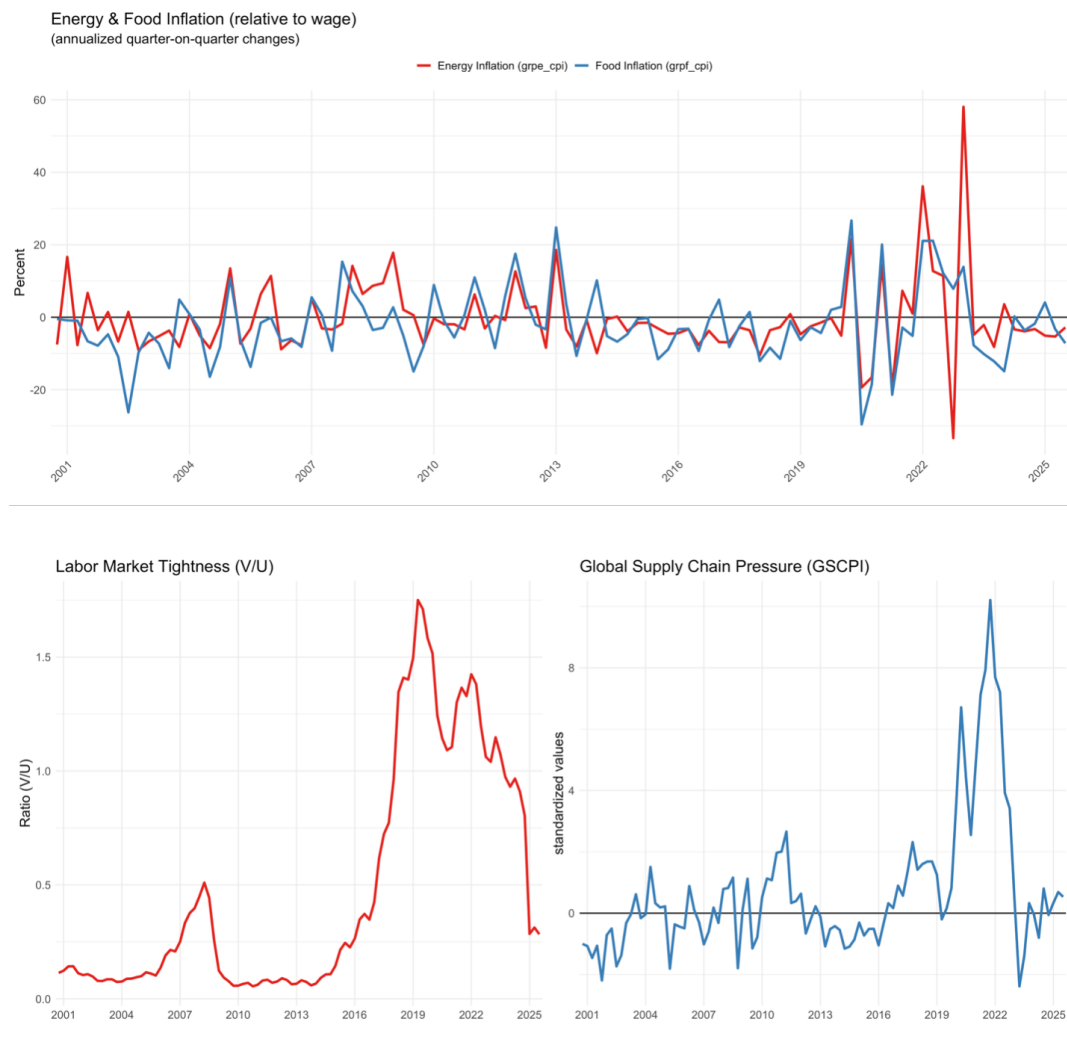


Figure 4.2: Development of Exogenous Variables

Figure 4.2 illustrates the sequence of external supply shocks and the evolution of domestic labor market conditions. The charts reveal a chronological succession of overlapping shocks. Initially, the economy experienced severe global supply chain bottlenecks, reflected in the sharp spike of the Global Supply Chain Pressure Index (GSCPI) during 2020 and 2021. This was subsequently followed by extreme positive shocks to relative energy prices and, with a slight lag, relative food prices. A notable and highly specific feature of the Czech data is the trajectory of the V/U ratio. The domestic labor market was already exceptionally tight in the pre-pandemic years. Following a brief moderation during the initial lockdowns, labor market tightness rapidly returned to historically high levels, providing a strong, continuous underlying environment for domestic wage pressures.

4.2.1 Energy price

Energy price inflation is computed as the quarter-on-quarter growth rate of the relative price of energy, measured as the rate of change of the ratio of the national CPI energy component to compensation per employee. The energy price data is not seasonally adjusted, as applying seasonal filters could artificially smooth out the true magnitude and exact timing of sudden, exogenous price shocks. Furthermore, utilizing the national CPI energy component rather than global commodity prices more accurately captures the actual cost shocks that are transmitted to the domestic economy.

4.2.2 Food prices

Food price inflation is computed as the annualized quarter-on-quarter growth rate of the relative price of food, measured as the rate of change of the ratio between the National CPI Food index and compensation per employee. Similar to energy prices, we use not seasonally adjusted food price data to avoid smoothing out the exact timing and magnitude of price shocks. Furthermore, utilizing the National CPI Food index rather than global agricultural commodity prices more accurately captures the actual retail cost shocks transmitted to the domestic economy. As highlighted by Bruha *et al.* (2020), domestic consumer food prices often decouple from raw agricultural producer prices due to fluctuations in retail margins and cyclical domestic demand pressures. Therefore, the national index inherently accounts for these local processing costs, distribution margins, and domestic market specifics.

4.2.3 Labor market tightness

Labor market tightness is primarily measured as the ratio of job vacancies to unplaced job seekers (the V/U ratio), using administrative data from Ministry of Labour and Social Affairs (MoLSA). In line with the framework developed by Blanchard & Bernanke (2023), the V/U ratio is preferred because it captures both labor demand and supply, whereas the unemployment rate represents only the supply side. This makes it a more effective metric for representing search and matching frictions, which is crucial for the post-pandemic period when labor markets were distorted by fluctuations in participation and various job retention schemes.

Furthermore, the choice of the V/U ratio is motivated by the evolution of

the domestic labor market. As noted by Unal & Maialeh (2024), the Czech Republic experienced a significant outward shift of the Beveridge curve after 2021, which aligns with findings in the US (Blanchard & Bernanke 2023). This structural shift implies that for any given level of unemployment, the number of vacancies was much higher than in the pre-pandemic period. This indicates a higher level of labor market pressure that would not be fully captured by the unemployment rate alone.

Unlike some economies (Ghomi *et al.* 2024), where the correlation between these two proxies remained stable, the structural shift in the Czech Republic makes the V/U ratio a more precise indicator of inflationary pressures originating in the labor market. Nevertheless, as a robustness check, we alternatively specify labor market tightness using the unemployment gap, defined as the deviation of the seasonally adjusted quarterly unemployment rate from its pre-pandemic level (2019Q4). These alternative results are presented in Table C.6 in the Appendix.

4.2.4 Shortage

Supply shortages are captured using the GSCPI, formulated as a quarterly mean standardized by its historical 1999–2019 baseline. In their original framework for the US, Blanchard & Bernanke (2023) utilized the Google Trends search intensity for the term "shortage" as a proxy for supply chain disruptions. Contrary to their approach, we instead rely solely on the GSCPI. Using Google search data for local language translations (such as "nedostatek" in Czech) yields highly noisy and unreliable results, as the term is overly broad and frequently captures internet searches entirely unrelated to macroeconomic supply bottlenecks.

4.2.5 Productivity

Productivity is measured as the annualized quarter-on-quarter growth rate of total economy labor productivity per person. Consistent with the original framework by Blanchard & Bernanke (2023), this series is transformed using an 8-quarter moving average. The moving average is preferred as it filters out the high-frequency noise and extreme volatility inherent in raw quarterly data. This makes it a better metric for representing underlying fundamental productivity trends, particularly during the pandemic and post-pandemic periods when short-term productivity measures were heavily distorted by sudden

compositional shifts in the workforce and the widespread use of job retention schemes.

4.2.6 Structural Price Gap

To construct the variables for our exploratory extension representing the structural maturity of the Czech economy, we utilize the Price Level Indices (PLIs) provided by Eurostat. We select the PLI based on Household Final Consumption Expenditure (HFCE) to ensure conceptual consistency with our dependent variable, consumer price inflation. The data is normalized such that the EU-27 average equals exactly 100 in each respective year.

Because official price level indices are published at an annual frequency, the series is transformed into quarterly data to match the frequency of the semi-structural framework. Given that price-level convergence is a slow-moving structural phenomenon, a cubic spline interpolation is applied to disaggregate annual observations into a smooth quarterly series (PLI_t). While this approach is standard for slow-moving structural variables, it may smooth short-term variation and potentially weaken the estimated interaction effects.

To express this metric as a "gap" representing the distance from the European core, we define:

$$GAP_t = 100 - PLI_t \quad (4.1)$$

A higher value of GAP_t therefore indicates a lower relative price level compared to the EU-27 average. This variable is used in the interaction terms to capture potential state-dependent amplification of external shocks.

To capture the direct, slow-moving inflationary pressure associated with the BS convergence effect, we construct a second variable, denoted as $GAPMA_t$, defined as an 8-quarter backward-looking moving average of the gap. This smoothed measure serves as a proxy for long-term convergence trends.

It is important to note that the price level index is inherently linked to cumulative inflation dynamics. As a result, the constructed GAP_t variable may partially reflect past inflation developments. While the use of lagged and smoothed versions of the gap mitigates contemporaneous feedback effects, it does not fully eliminate potential endogeneity.

Therefore, the GAP variable should be interpreted as a proxy for the broader structural environment rather than as a strictly exogenous measure of price-level differences.

Chapter 5

Empirical Methodology

Before specifying the structural equations, it is necessary to establish the time-series properties of the dataset to prevent spurious regression. Table B.1 in the Appendix summarizes the Augmented Dickey-Fuller (ADF) unit root diagnostics for all variables.

The results indicate that the core endogenous variables in their growth-rate forms, such as nominal wage growth (Δw_t) and consumer price inflation (Δp_t), are stationary at the 1% significance level. Crucially, the external supply shock variables, including the growth rates of relative energy and food prices (ΔRPE and ΔRPF), also exhibit stationarity. While the p-value for relative food price growth stands at 0.059 (stationary at the 10% significance level). Variables that exhibit higher persistence, such as the labor market tightness ratio (V/U) and the structural price level gap (GAP_t), are maintained in the system without differencing due to their theoretically bounded nature.

5.1 The Wage Equation

The wage equation models the dynamic evolution of nominal wage growth. We assume that wage growth (Δw_t) depends on its own lagged values, past inflation expectations, and the state of the labor market. Furthermore, to account for the extraordinary labor market disruptions caused by the COVID-19 pandemic, dummy variables are included for the lockdown periods in 2020Q2 and 2021Q1. The empirical specification takes the following form:

$$\begin{aligned} \Delta w_t = & \alpha_0 + \sum_{i=1}^4 \alpha_{1,i} \Delta w_{t-i} + \sum_{i=0}^4 \alpha_{2,i} CF1_{t-i} + \sum_{i=0}^4 \alpha_{3,i} \left(\frac{V}{U} \right)_{t-i} \\ & + \alpha_4 CATCH_UP_t + \alpha_5 \Delta gpty_t + \delta D_t + \varepsilon_{1,t} \end{aligned} \quad (5.1)$$

subject to the long-run homogeneity restriction:

$$\sum_{i=1}^4 \alpha_{1,i} + \sum_{i=0}^4 \alpha_{2,i} = 1 \quad (5.2)$$

Here, $CF1$ represents short-term inflation expectations, V/U captures labor market tightness, $CATCH_UP_t$ measures the realized inflation surprise (the gap between past actual inflation and past expectations), and $\Delta gpty_t$ accounts for the labor productivity trend. The vector D_t contains the pandemic dummy variables.

5.2 The Price Equation

The price equation captures the cost-push pricing behavior of firms. Consumer price inflation (Δp_t) is driven by past inflation, wage costs, and external supply shocks relative to wages. Furthermore, a specific policy variable is introduced to account for the government's administrative interventions aimed at shielding households from extreme energy price spikes. This variable captures the Energy Savings Tariff and the subsequent energy price cap. The equation is estimated as:

$$\begin{aligned} \Delta p_t = & \beta_0 + \sum_{i=1}^4 \beta_{1,i} \Delta p_{t-i} + \sum_{i=0}^4 \beta_{2,i} \Delta w_{t-i} \\ & + \sum_{i=0}^4 \beta_{3,i} \Delta RPE_{t-i} + \sum_{i=0}^4 \beta_{4,i} \Delta RPF_{t-i} \\ & + \sum_{i=0}^4 \beta_{5,i} SHORTAGE_{t-i} + \beta_6 \Delta gpty_t + \theta D_t^{cap} + \varepsilon_{2,t} \end{aligned} \quad (5.3)$$

subject to the restriction:

$$\sum_{i=1}^4 \beta_{1,i} + \sum_{i=0}^4 \beta_{2,i} = 1 \quad (5.4)$$

In this specification, ΔRPE and ΔRPF denote the growth rates of relative energy and food prices, respectively. The *SHORTAGE* variable represents global supply chain bottlenecks. To accurately reflect the unique policy environment in the Czech Republic, D_t^{cap} is constructed as a continuous variable rather than a simple binary dummy. It captures the direct effect of government interventions on energy prices. The variable is set to 1 in 2022Q4 because the introduction of the “savings tariff” temporarily reduced households’ electricity expenses, causing an artificial drop in reported consumer prices. In 2023Q1, the variable is set to -1 . The savings tariff ended and was replaced by energy price ceilings (Czech Statistical Office 2023). As a result, households faced a sudden jump in costs. This created a statistical baseline effect, where the reported prices rebounded. The homogeneity restriction ensures that, in the absence of productivity growth or external commodity shocks, an equal permanent increase in wage growth and past inflation will eventually raise current price inflation by the exact same amount, maintaining a constant real wage mark-up.

5.3 The Price Level Extension

To empirically assess the theoretical hypothesis developed in Chapter 3, we augment the baseline price equation (5.3). We follow a state-dependent specification approach, modeling the pass-through coefficients of both energy and food shocks as linear functions of the domestic price level gap (*GAP*).

The extended empirical price equation takes the following form:

$$\begin{aligned}
 \Delta p_t = & \beta_0 + \sum_{i=1}^4 \beta_{1,i} \Delta p_{t-i} + \sum_{i=0}^4 \beta_{2,i} \Delta w_{t-i} \\
 & + \sum_{i=0}^4 \left(\beta_{3,i} + \gamma_{3,i} GAP_{t-1-i} \right) \Delta RPE_{t-i} \\
 & + \sum_{i=0}^4 \left(\beta_{4,i} + \gamma_{4,i} GAP_{t-1-i} \right) \Delta RPF_{t-i} \\
 & + \sum_{i=0}^4 \beta_{5,i} SHORTAGE_{t-i} + \beta_6 \Delta gpty_t \\
 & + \phi GAPMA_t + \theta D_t^{cap} + \varepsilon_{2,t}
 \end{aligned} \tag{5.5}$$

subject to the identical long-run homogeneity restriction:

$$\sum_{i=1}^4 \beta_{1,i} + \sum_{i=0}^4 \beta_{2,i} = 1 \quad (5.6)$$

In this extended specification, the pass-through of global commodity shocks is no longer assumed to be uniform. The terms $(\beta_{3,i} + \gamma_{3,i}GAP_{t-1-i})$ and $(\beta_{4,i} + \gamma_{4,i}GAP_{t-1-i})$ are designed to capture potential state-dependent variation in the pass-through of energy and food shocks, respectively. We hypothesize that $\gamma_{3,i} > 0$ and $\gamma_{4,i} > 0$, implying that a larger price level gap (a relatively cheaper economy) acts as an amplifier for external commodity shocks.

To ensure an intuitive interpretation of the multiplier effect, the structural state of the economy is quantified using the PLI. We construct the structural gap as the inverse distance from the European core:

$$GAP_t = 100 - PLI_t \quad (5.7)$$

where PLI_t is the official harmonized index normalized such that the EU-27 average equals 100. Consequently, a higher value of GAP_t indicates a lower domestic price level.

Crucially, we utilize the one-quarter lagged value of this gap (GAP_{t-1-i}) in the interaction terms to mitigate contemporaneous feedback effects. Nonetheless, since the price level is inherently linked to cumulative inflation dynamics, lagging the variable does not fully eliminate potential endogeneity. Therefore, the estimated interaction effects should be interpreted as conditional correlations.

We explicitly adopt a linear functional form for the state-dependent interaction. This methodological choice is driven by several factors. First, in the absence of a definitive theoretical consensus on the exact shape of the vulnerability function, the linear specification serves as a robust, first-order Taylor approximation commonly utilized in Time-Varying Parameter (TVP) models. Second, it avoids imposing unverified assumptions about convexity. Finally, the linear form provides highly interpretable coefficients, allowing us to quantify the exact reduction in shock pass-through for every percentage point of price-level convergence.

Furthermore, we set up the variable $GAPMA_t$ to capture the direct, slow-

moving structural inflation arising from the BS convergence effect. To reflect this long-term macroeconomic trend and filter out short-term cyclical noise, $GAPMA_t$ is constructed as an 8-quarter backward-looking moving average of the underlying gap. Including the convergence term helps isolate the interaction coefficients ($\gamma_{3,i}$ and $\gamma_{4,i}$) from underlying convergence trends.

Importantly, the introduction of these state-dependent extensions does not require any additional parameter restrictions. Because the supply shocks (ΔRPE and ΔRPF) are stationary relative price changes that converge to zero in the steady state, the non-linear interaction terms dictate only the transitory dynamic adjustments. The original restriction ($\sum \beta_{1,i} + \sum \beta_{2,i} = 1$) remains both necessary and sufficient to ensure the long-run nominal neutrality of the model, maintaining a constant real wage mark-up in the absence of productivity growth or commodity shocks.

5.4 The Inflation Expectations Equations

To close the model, the dynamics of inflation expectations are explicitly quantified. Short-term inflation expectations ($CF1_t$) are assumed to depend on their own past values, long-term inflation expectations ($CF3_t$), and actual realized inflation.

$$CF1_t = \sum_{i=1}^4 \gamma_{1,i} CF1_{t-i} + \sum_{i=0}^4 \gamma_{2,i} CF3_{t-i} + \sum_{i=0}^4 \gamma_{3,i} \Delta p_{t-i} + \varepsilon_{3,t} \quad (5.8)$$

subject to the restriction:

$$\sum_{i=1}^4 \gamma_{1,i} + \sum_{i=0}^4 \gamma_{2,i} + \sum_{i=0}^4 \gamma_{3,i} = 1 \quad (5.9)$$

Both explanatory variables are expected to exert a positive influence on current short-term expectations. This equation allows us to test whether expectations are strongly anchored. If they are well-anchored, the coefficients on long-term expectations ($\gamma_{2,i}$) should dominate, and short-term expectations should respond less to past movements in actual inflation.

Finally, long-term inflation expectations ($CF3_t$) are modeled similarly, depending on their own lags and actual inflation:

$$CF3_t = \sum_{i=1}^4 \delta_{1,i} CF3_{t-i} + \sum_{i=0}^4 \delta_{2,i} \Delta p_{t-i} + \varepsilon_{4,t} \quad (5.10)$$

subject to:

$$\sum_{i=1}^4 \delta_{1,i} + \sum_{i=0}^4 \delta_{2,i} = 1 \quad (5.11)$$

This last equation acts as the anchor of the system. It helps to determine how credible the central bank's inflation target remains in the eyes of financial markets following an aggregate price shock.

To summarize the methodological choices made in constructing the system of equations, Table 5.1 compares the equation specifications of the original U.S. framework with the model adapted for the Czech Republic in this thesis.

Table 5.1: Comparison of Model Adaptations: Equation Specifications and Extensions

Equation	Original BB Model (US)	Czech Adaptation (CZ)
Wage equation –		Included dummy variables for lockdowns in 2020Q2 and 2021Q1.
Price equation –		Included a continuous dummy variable for government energy interventions (savings tariff 2022Q4 = 1, price cap 2023Q1 = -1).
Price equation –		Added a price level gap defined as $100 - PLI_t$ to capture structural convergence.

Chapter 6

Empirical Results

This chapter presents the estimation results using the full data sample from 2000Q4 to 2025Q3. We use the complete dataset rather than restricting the estimation to the pre-pandemic period for three main reasons. First, consistent macroeconomic data for the Czech Republic are only available from 2000. Relying solely on pre-pandemic data would result in a relatively short time series. Second, while recent literature (Bernanke & Blanchard 2024) estimated models using data that ended during the peak of the high-inflation episode, our sample includes the data after it. Third, key variables such as the supply shortage index showed almost no variation before 2020. Using the full sample is therefore necessary to properly identify their structural impact on inflation.

6.1 Baseline Model Results

This section presents the results of the model. Coefficient sums are taken from the restricted specification to stay consistent with the structural assumptions in Blanchard & Bernanke (2023). Statistical significance is evaluated using the unrestricted model because it provides a more conservative and robust assessment. In this way, the point estimates follow the theoretical structure, while the significance is not influenced by the imposed restrictions. Using p-values from the restricted model could artificially increase significance, which would create a risk of overfitting or misleading conclusions about the importance of the coefficients.

Before discussing the individual equations, it is important to address the long-run homogeneity restrictions. Formal Wald tests (reported in Table B.2 in the Appendix) often reject these restrictions in our data. Wald tests can

exhibit size distortions in finite samples with multiple equations, leading to the over-rejection of theoretically valid long-run constraints (Bera *et al.* 1981). This behavior is often driven by empirical limitations such as short data samples, multicollinearity, or measurement errors.

Therefore, the restriction is treated as a structural identifying assumption rather than a statistically testable hypothesis. Following (Bernanke & Blanchard 2024), we retain the restrictions because they provide the long-run equilibrium structure necessary for a meaningful economic interpretation and ensure comparability with the BB framework. The restricted model thus serves as a theoretical benchmark.

6.1.1 Wage Equation

First, we estimate the wage equation. The results are presented in Table 6.1. The long-run effects of the explanatory variables are represented by the sum of their estimated coefficients (including contemporaneous and lagged values), which are reported in the column labeled Sum (Σ). In addition, we test the statistical significance of these long-run coefficients using two p-values. The first p-value tests the null hypothesis that the sum of the coefficients is zero. The second p-value tests the joint hypothesis that all individual coefficients for a given variable are equal to zero.

Based on the restricted model, nominal wage growth is primarily driven by forward-looking inflation expectations (0.66), consistent with evidence for the EA (Bernanke & Blanchard 2024). Regarding the wage growth equation, the catch-up term has a statistically significant effect, suggesting that workers were not fully compensated for unexpectedly high inflation¹. Interestingly, this is the opposite of findings in other countries and indicates that Czech households do not successfully demand compensation for past unexpected inflation. This aligns with reports by ING (2024) that nominal wage growth already slowed by mid-2024, despite the massive drop in real wages during the inflation shock in 2022–2023. This may be explained by the lack of centralized union bargaining and the absence of automatic wage indexation in the Czech Republic. Instead, wage growth depends heavily on the cash flow of individual firms. In the face of cost-push inflation, financially constrained firms tend to freeze base wages or cut flexible components, such as bonuses (Babecký *et al.* 2017).

¹While the coefficients for the third and fourth lag are positive (though statistically insignificant), the coefficient on the second lag is negative and significant at the 10% level.

Table 6.1: Estimation Results for the Wage Equation

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
<i>Dependent variable: Wage growth (Δw_t)</i>					
Wage growth (Δw_t)		0.34	0.09	0.00	1 to 4
Short-term exp. ($CF1_t$)		0.66	0.21	0.02	1 to 4
Labor tightness (V/U)		2.92	0.00	0.00	1 to 4
Catch-up term ($CATCHUP_t$)		-1.42	0.07	0.01	1 to 4
Productivity trend ($\Delta gpty_t$)		0.24	0.23	0.23	1
R^2		0.612			

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted model. The sum of coefficients (Σ) is estimated from the restricted model, satisfying the theoretical homogeneity condition $\Sigma\Delta w + \Sigma CF1 = 1$.

Regarding labor market tightness, we observe a large and statistically significant positive coefficient (2.92). This implies that a tighter labor market translates into strong wage growth pressures. This result reflects the highly rigid nature of the Czech labor market, where job vacancies exceeded the number of applicants even before the pandemic. These findings are consistent with Sirot *et al.* (2024), who show that labor market tightness passes through to inflation gradually but very strongly².

For the productivity trend, the estimated coefficient is positive (0.24) but statistically insignificant. This result corresponds with findings for most EU countries, where the effect of productivity on wage growth is also typically positive yet statistically insignificant (Bernanke & Blanchard 2024). Finally, the lockdown dummies are negative and significant for the assigned quarters.

6.1.2 Price Equation

Regarding the price equation (Table 6.2), the model explains a substantial share of inflation dynamics (R-squared of 0.92). The imposed restriction implies that,

²Specifically, they find that a one-standard-deviation shock to labor market tightness leads to a cumulative increase in inflation of approximately 3 percentage points over a three-year horizon, regardless of monetary policy or exchange rate movements.

in the long run, increases in nominal wages and past prices are fully passed through into final prices.

Within the restricted specification, the estimated wage pass-through is 0.26, which is relatively moderate compared to the high inflation persistence of 0.74. Both effects are jointly statistically significant. These values are broadly in line with findings for European economies, but contrast with the US, where the wage channel is typically stronger and inflation persistence lower.

Overall, the results suggest that price-setting in the Czech economy is relatively backward-looking, with firms placing greater weight on past inflation dynamics than on contemporaneous labor cost pressures.

Table 6.2: Estimation Results for the Price Equation

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
<i>Dependent variable: Price growth (Δp_t)</i>					
Price growth (Δp_t)		0.74	0.17	0.00	1 to 4
Wage growth (Δw_t)		0.26	0.11	0.00	0 to 4
Relative energy price (ΔRPE_t)		0.09	0.04	0.00	0 to 4
Relative food price (ΔRPF_t)		0.03	0.31	0.00	0 to 4
Supply shortages ($SHORTAGE_t$)		0.19	0.00	0.00	0 to 4
Productivity trend ($\Delta gpty_t$)		-0.03	0.74	0.74	1
R^2		0.922			

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted model. The sum of coefficients (Σ) is estimated from the restricted model, satisfying the theoretical homogeneity condition $\Sigma\Delta p + \Sigma\Delta w = 1$.

Regarding cost-push shocks, they exhibit joint statistical significance. Notably, the long-term impact of energy prices is approximately three times larger than that of food prices. The calculated long-term multiplier is 35% for energy prices and 12% for food prices³. For energy, the multiplier significantly exceeds its weight in the 2025 CPI basket (25.8%)⁴, indicating substantial positive

³The multiplier would be calculated, for example for energy, as $0.09/(1-0.74)=0.35$

⁴Note that this category includes imputed rentals; see Chapter 5.

indirect effects as energy costs spill over into the prices of other goods and services. Conversely, the multiplier for food (12%) is lower than its basket weight (17.7% (CSU 2025)). This suggests a negative indirect effect, implying that firms might have absorbed higher food costs through compressed profit margins or that these costs led to reduced demand in other sectors of the economy. This pattern differs from many other European countries, where the impact of food prices on inflation has typically been more dominant than the impact of energy prices (Bernanke & Blanchard 2024).

Another significant factor is supply shortages (0.19), which are statistically significant. As a small open economy, the Czech Republic was particularly vulnerable to these shocks. Theoretically, productivity growth is expected to have a disinflationary (negative) effect on prices. However, our estimate for the Czech Republic (-0.03) is statistically insignificant. This result is consistent with other European countries (Bernanke & Blanchard 2024), in contrast to the US, where productivity exhibits a significant disinflationary impact (-0.14). In the European context, these productivity effects generally remain statistically insignificant.

6.1.3 Inflation Expectations

The results for the short-term inflation expectations equation ($CF1$) are presented in Table 6.3. The model has a high explanatory power, with an R^2 of 0.981. To ensure theoretical consistency, we imposed the homogeneity restriction, assuming that the sum of coefficients for inflation expectations and price growth equals one.

The imposed restriction ensures theoretical consistency, reflecting the absence of long-run money illusion in price-setting; ultimately, short-term expectations fully adjust to reflect realized inflation and long-term anchors.

Table 6.3: Estimation Results for Short-Term Inflation Expectations ($CF1$)

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
Short-term exp. ($CF1_t$)		0.850	0.000	0.000	1 to 4
Long-term exp. ($CF3_t$)		0.073	0.271	0.349	0 to 4
Price growth (Δp_t)		0.076	0.036	0.000	0 to 4
R^2		0.981	<i>Restriction: $\Sigma CF1 + \Sigma CF3 + \Sigma \Delta p = 1$</i>		

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted model. The sums of coefficients (Σ) are estimated from the restricted model satisfying the theoretical condition of nominal neutrality.

The substantially larger coefficient for short-term expectations (0.850) compared to past price growth (0.076) indicates that short-term inflation expectations are primarily driven by their own past values and long-term anchors rather than by recent realized inflation. This suggests that expectation formation is forward-looking, with limited sensitivity to short-term price fluctuations. The limited role of past inflation suggests that expectations remain reasonably well-anchored. As a result, short-term price shocks do not fully pass through to wage demands, reducing the risk of a persistent inflationary spiral.

Table 6.4: Estimation Results for Long-Term Inflation Expectations ($CF3$)

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
Long-term exp. ($CF3_t$)		0.994	0.000	0.000	1 to 4
Price growth (Δp_t)		0.006	0.036	0.000	0 to 4
R^2		0.926	<i>Restriction: $\Sigma CF3 + \Sigma \Delta p = 1$</i>		

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted model.

Long-term inflation expectations (Table 6.4) are modeled following Menz (2024), with a homogeneity restriction imposed to ensure comparability and assess expectation anchoring. The results show that past long-term expectations dominate (0.994), while the response to realized inflation is negligible

(0.006), consistent with well-anchored expectations and sustained central bank credibility.

6.1.4 Stability of results

We use a Chow predictive test to assess whether the pandemic period was associated with structural changes in the estimated relationships. The results indicate statistically significant structural breaks in the wage, price, and short-term expectations equations, consistent with findings for the aggregate EA (Bernanke & Blanchard 2024).

These results imply that the Ordinary Least Squares (OLS) coefficients estimated over the full sample should be interpreted as averages across potentially different regimes. The identified breaks may reflect changes in price-setting and wage-bargaining behavior during the high-inflation period, as well as possible non-linearities in the transmission of cost shocks.

Interestingly, while short-term expectations exhibit evidence of structural change, long-term expectations appear relatively stable. This pattern is consistent with the notion that long-term expectations remained anchored to the inflation target, although this conclusion should be interpreted with caution given the presence of structural breaks in other parts of the system.

As a result, the impulse response functions and historical decompositions presented in Section 6.2 reflect full-sample average dynamics and may not precisely capture the post-pandemic transmission regime.

6.2 System Dynamics

6.2.1 Impulse Response Functions (IRFs)

To quantify the overall dynamic effects of exogenous supply-side and labor market shocks, we compute IRFs⁵. The semi-structural nature of the model allows us to capture not only the direct impacts of these shocks but also the indirect second-round feedback effects operating through the wage-price spiral. Following Blanchard & Bernanke (2023), the magnitude of each exogenous shock is calibrated to one standard deviation, calculated during the crisis and post-pandemic period (2020Q1–2023Q3). This ensures that the simulations accurately reflect the scale of the recent inflationary episode. We differenti-

⁵IRF display the unadjusted sum of the annualized quarterly responses

ate between two types of shocks based on their persistence. Shocks to relative energy prices, relative food prices, supply shortages, and productivity are modeled as temporary. They increase exogenously by 1 Standard Deviation (SD) in the first quarter and immediately return to zero. Their prolonged macroeconomic impact is therefore entirely driven by the system's internal transmission mechanisms. In contrast, the shock to labor market tightness (V/U ratio) is treated as permanent, remaining elevated throughout the simulation. This reflects the reality that structural changes in the labor market require more time to resolve.

For clarity of interpretation, the IRFs for wage and price inflation are presented as cumulative responses. The cumulative value of these variables converts their growth rates into aggregate price and wage levels. This approach eliminates short-term volatility of wage inflation and clearly illustrates the long-term permanent impact of shocks on the economy. In contrast, the IRFs for short- and long-term inflation expectations are reported in standard levels.

Figure 6.1 shows the cumulative IRFs for nominal wages, illustrating the long-run effect of structural shocks on the aggregate wage level.

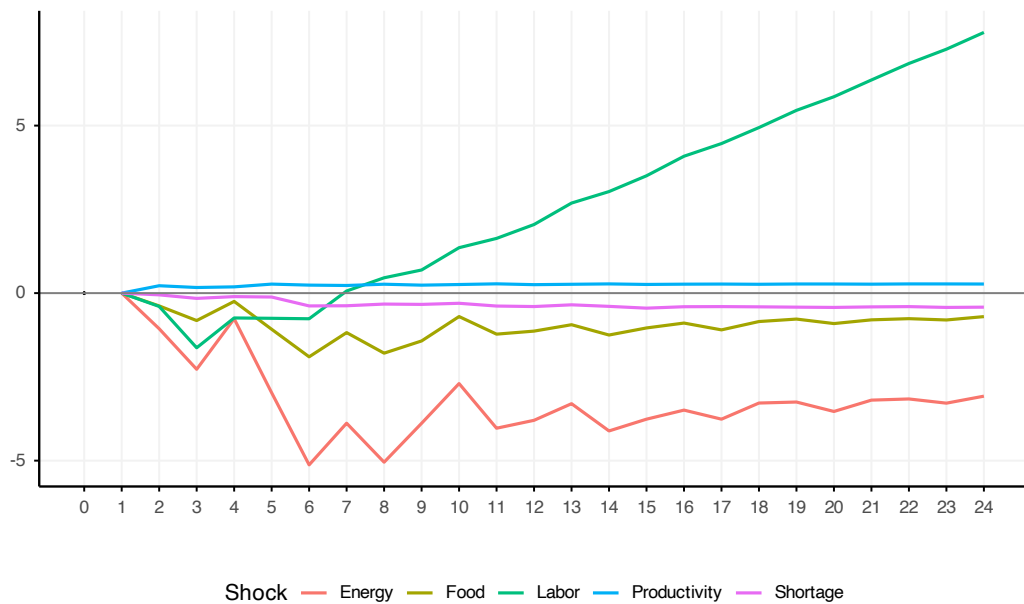


Figure 6.1: Impulse Response Function of Wage growth

An increase in relative energy prices directly increases consumer inflation, which then pushes up wage growth through higher inflation expectations. The cumulative IRF shows that the overall long-run effect of the energy shock on

the nominal wage level is actually negative, reducing it by about 4% compared to the baseline.

These findings are consistent with Nikiforos & Grothe (2023), who argue that foreign price shocks have strong contractionary effects. A sharp increase in energy prices acts as a transfer of income from domestic consumers to foreign producers and from labor to capital. Since workers typically have a high propensity to consume, this redistribution of income leads to a significant drop in aggregate demand. This mechanism is particularly relevant for the Czech Republic, which is characterized by a significant gap between GDP and Gross National Income (GNI) due to high dividend outflows to foreign owners (Dražalová 2025).

The food price shock has a similar negative effect, though it is smaller than the energy shock. Supply shortages also lead to a negative response, but the impact is negligible.

Therefore our results suggest that this negative pressure from the cooling economy outweighs the workers' push for inflation compensation. Consequently, the nominal wage level ends up lower in the long run.

Finally, the productivity shock is positive, but its overall effect on the wage level is very small. In contrast, the labor market shock works differently. After a small negative response at the beginning, a permanently tighter labor market causes a positive and steadily increasing cumulative wage level. This linear rise is expected, as the labor market shock is modeled as permanent.

Figure 6.2 shows the cumulative IRFs for the price level.

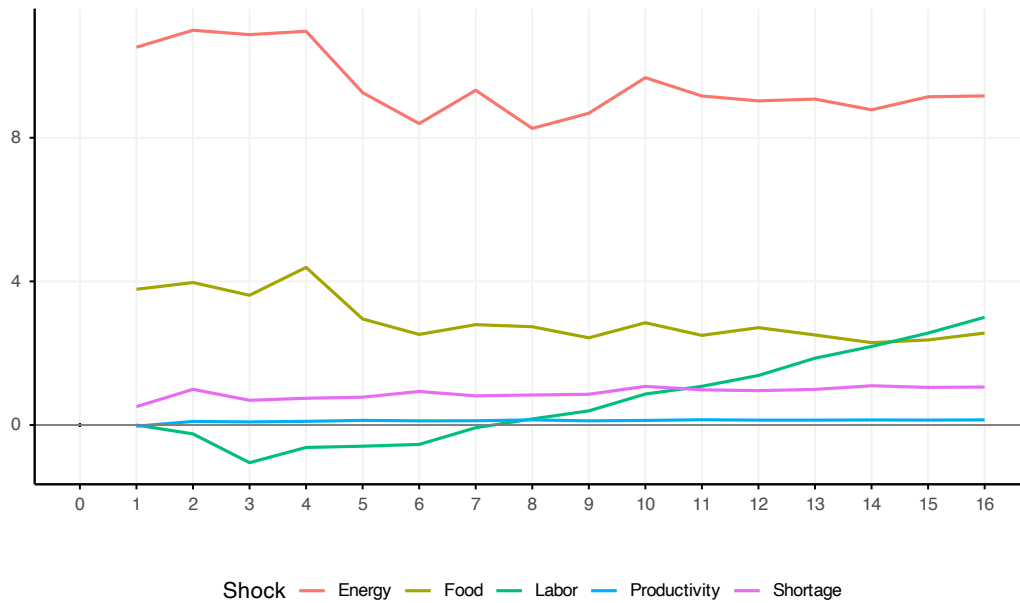


Figure 6.2: Impulse Response Function of Inflation

For energy and food prices, we see an immediate jump in the price level. A one-standard-deviation shock to energy prices raises the price level by about 10%, while food prices increase it by about 4%. After this initial jump, the price level remains constant or declines slightly. This horizontal line shows that the shocks are transitory. They cause a one-time permanent shift in prices, but they do not lead to further inflation after the shock ends.

Supply bottlenecks have a similar but smaller effect, raising the price level by around 0.5% to 1%. These results indicate that the high inflation after the pandemic was mainly driven by these external commodity and supply shocks.

In contrast, the labor market shock shows a different pattern. While the initial impact is small and negative, after year and half the price level continues to rise linearly over time. Because the labor market tightening is modeled as a permanent structural change, it creates a persistent and long-lasting upward pressure on prices. This suggests that while commodity shocks caused the initial jump in prices, the tight labor market keeps inflation higher for a longer period.

The IRFs for short-term inflation expectations show that energy and food price shocks have a strong but temporary impact. An energy price shock increases expectations by a maximum of 1.5 percentage points, while a food

price shock raises them by 0.5 percentage points. Both shocks peak after two quarters. After the peak, they decrease sharply.

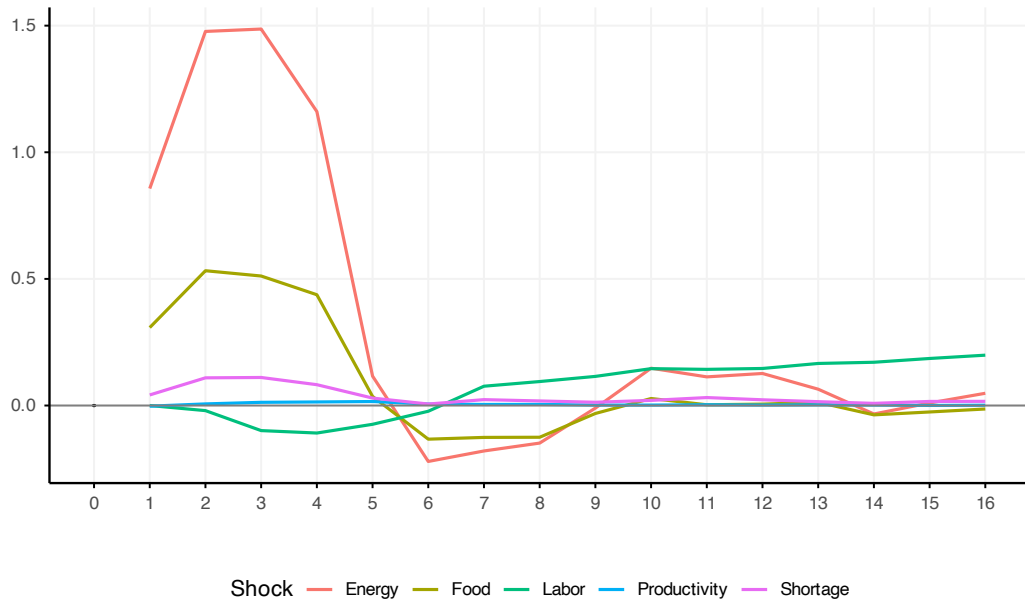


Figure 6.3: IRF of Short-term inflation expectations

In contrast, the labor market shock has a much more gradual and persistent effect. Although the initial impact is small and negative, expectations rise steadily over time, reaching 0.2 percentage points. This reflects the long-lasting nature of structural changes in the labor market. Finally, the effects of productivity shocks and supply shortages are negligible.

As shown in Figure 6.4, the response of long-term inflation expectations follows a similar initial pattern to short-term expectations, but with much higher persistence. While the magnitude of the shocks is significantly smaller.

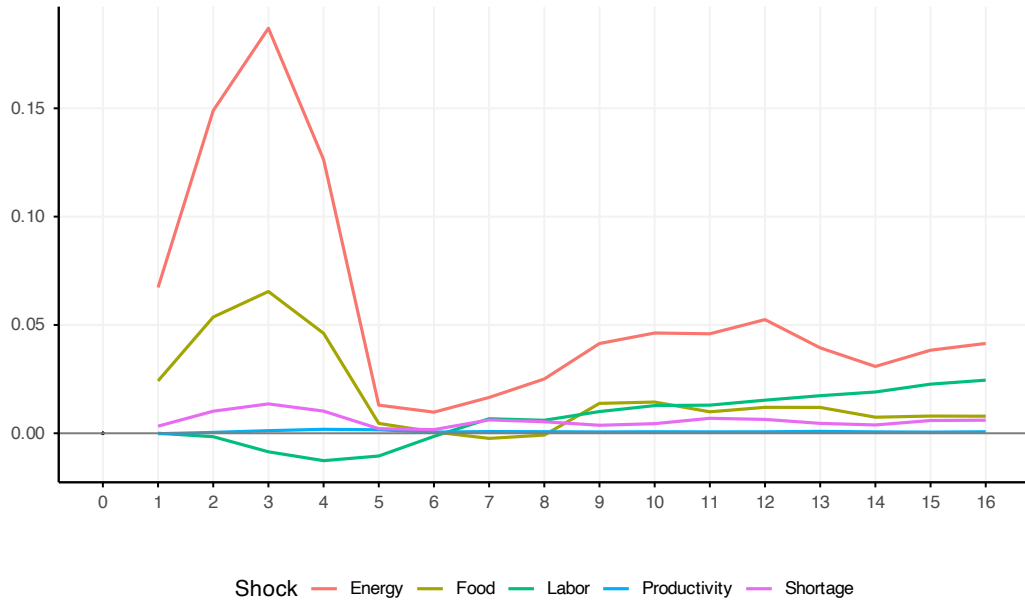


Figure 6.4: IRF of Long-term inflation expectations

An energy price shock increases long-term expectations by nearly 0.2 percentage points, and food price shocks lead to a peak of approximately 0.06 percentage points. Both shocks reach their peak in the second quarter. Although there is a sharp decline after the peak, the expectations remain positive and do not return to zero. This suggests that large commodity shocks can lead to a slight unanchoring of long-term inflation expectation, but only of a very small magnitude.

The labor market shock shows a gradual and persistent upward trend. After a very small initial dip, it rises steadily to about 0.025 percentage points by the end of the simulation. This indicates that structural tightening in the labor market creates a small but permanent increase in long-run inflation beliefs. Finally, the impacts of productivity shocks and supply shortages remain negligible.

6.2.2 Historical Decomposition of post-pandemic developments

To evaluate the post-pandemic developments in inflation, wages, and expectations, we perform a historical decomposition of the full model. The analysis covers the period 5 years back up to 2025Q3 and accounts for the endogenous feedbacks between all variables in the system.

The shocks in the decomposition are constructed based on specific benchmarks to represent their relative contributions. Energy and food price shocks are modeled as deviations from zero, representing a counterfactual scenario where these shocks did not occur. In contrast, both the supply shortages and the labor market shock are defined as deviations from their pre-pandemic levels, fixed at the 2019Q4 values. These elements, together with estimated constants and pre-pandemic dynamics, form the "initial conditions" term.

Furthermore, in constructing the historical decompositions, specific care was taken to allocate policy and pandemic dummy variables to their appropriate economic components. The continuous policy variable capturing the energy savings tariff and price cap was aggregated directly into the energy component, as it constitutes a direct administrative intervention in energy pricing.

Regarding the treatment of pandemic lockdowns, we consider three different methodological approaches. The first approach, used in the original US model (Blanchard & Bernanke 2023), treats the lockdown dummy as a temporary data distortion and absorbs it into the residuals. This approach argues that lockdowns were temporary noise rather than fundamental macroeconomic shocks.

The second approach, consistent with Bounajm *et al.* (2024), incorporates the lockdown effect into the supply shortage component. The economic rationale is that mobility restrictions prevented consumers from spending on services, forcing a sudden shift in demand toward durable goods. This shift generated excess demand and logistical bottlenecks, making the lockdowns a primary trigger for global supply chain disruptions.

Finally, we adopt a third alternative, which isolates the lockdown effect as a standalone structural component, following the methodology of Menz (2024). This allows us to explicitly quantify the impact of government-imposed restrictions on economic activity separately from other supply-side factors.

Figures 6.5 and 6.6 show the historical decomposition for annual wage growth and price inflation. To improve readability, we transformed the annualized quarterly growth rates into annual rates.

For wages in the absence of shocks, the results indicate that nominal wages in the Czech Republic would have grown by approximately 8% (initial conditions). This high steady-state growth reflects the extreme tightness of the Czech labor market existing immediately before the pandemic. The decomposition shows that lockdowns in 2020Q2 and 2021Q2 caused a massive temporary drop in wage growth, reducing it by as much as 12 percentage points.

The labor market shock has had a negative contribution to wage growth since the end of 2021. Because the labor market is benchmarked to the peak tightness of 2019Q4, any subsequent cooling of the V/U ratio acts as a drag on wage growth relative to that baseline. Supply shortages and commodity prices (energy and food) had only a limited and fluctuating impact on wages compared to the domestic factors.

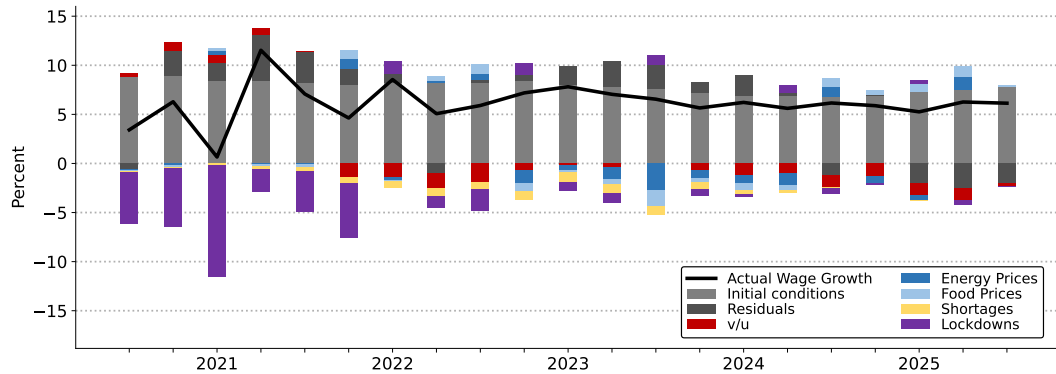


Figure 6.5: Historical Decomposition of Wage growth

Regarding price inflation (Figure 6.6), the results indicate that the surge following the Russian invasion of Ukraine was primarily driven by external supply-side factors rather than domestic demand.

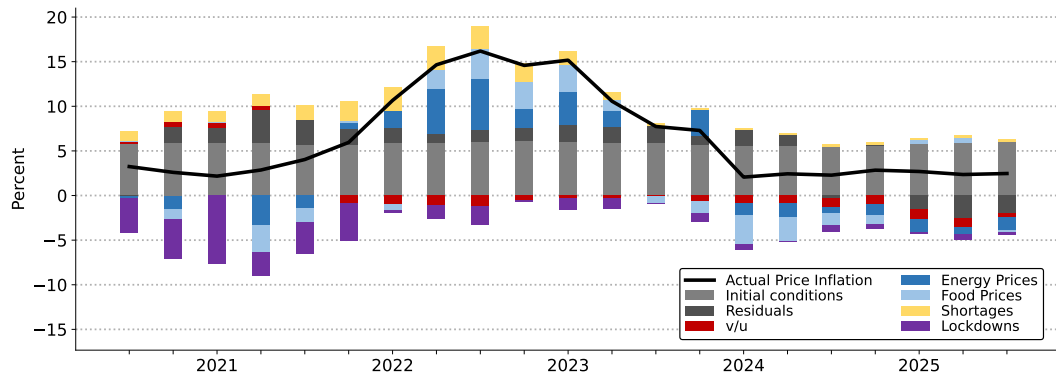


Figure 6.6: Historical Decomposition of Inflation

The initial conditions contribute approximately 5% to the annual inflation rate. While this exceeds the 2% inflation target. Before the pandemic, the Czech economy was characterized by significant overheating and a highly tight labor market, with inflation already reaching 3.7% in early 2020. The model captures this pre-existing momentum and structural tightness within the initial conditions component. During the early stages of the pandemic, the

combination of lockdowns and movements in energy and food prices exerted disinflationary pressure. These negative shocks temporarily suppressed price growth before the onset of the global commodity crisis.

The spike in inflation starting in late 2021 was driven largely by a combination of energy and food price shocks. At the peak in 2022, when inflation reached nearly 20%, these surging commodity costs, along with supply shortages reflecting global logistical bottlenecks, were the dominant factors. In 2024, energy and food prices acted as a cooling factor on inflation, as they decreased from their exceptionally high levels.

In contrast, the direct contributions of labor market shocks to price inflation were negligible and negative. Overall, the decomposition confirms that the Czech inflation peak was the result of a sequence of external commodity shocks and supply-chain disruptions.

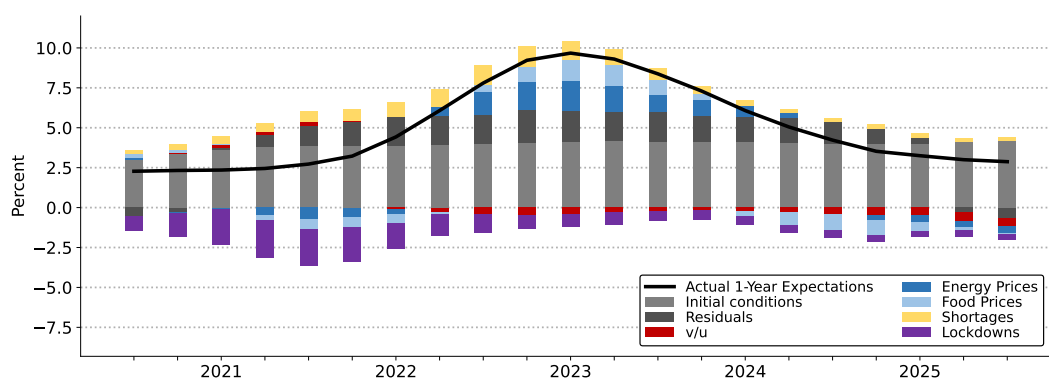


Figure 6.7: Historical Decomposition of Short-term inflation expectations

The decomposition of short-term inflation expectations (Figure 6.7) shows that initial conditions start at 3% and gradually rise to 4%. Since the beginning of the pandemic, lockdowns have exerted downward pressure on inflation expectations. A similar disinflationary effect was initially observed for commodity prices (energy and food). After the beginning of the war in Ukraine, these factors significantly pushed expectations upward. Throughout the observed period, supply shortages consistently contributed to higher expectations, accounting for approximately 1% in 2022. These effects began to fade around 2024, with energy and food prices eventually pulling expectations lower.

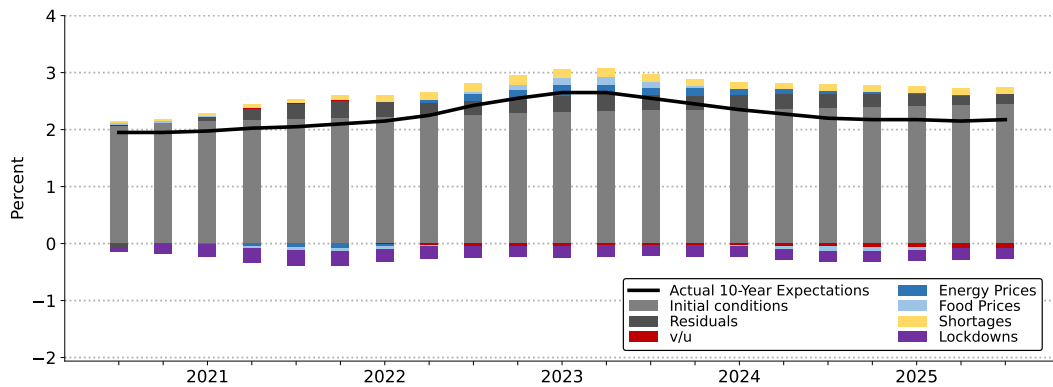


Figure 6.8: Historical Decomposition of Long-term inflation expectations

Furthermore, a large positive residual is visible from the start of the war until 2025. This raises the question of whether households expect higher inflation due to the lower price level in the Czech Republic relative to Western countries. Lastly, the labor market has contributed to lowering inflation expectations since 2022.

Regarding the decomposition of long-term inflation expectations (Figure 6.8), the results suggest that expectations remained relatively well-anchored. Over the long horizon, there was only a slight increase, with a peak of 3% in 2023. This increase is likely due to the pass-through of persistent cost shocks into three-year inflation expectations. Similar to the short-term decomposition, a positive residual is present. Conversely, lockdowns and labor market conditions exerted downward pressure on long-term inflation expectations.

6.3 Model Extension: The Role of the Price Level Gap

In this section, I extend the baseline price equation by incorporating a GAP variable. This extension examines whether the relative domestic price level is associated with the transmission of external shocks into domestic inflation.

6.3.1 Extended Price Equation Estimates

In Table 6.5 we can see interaction terms between commodity shocks and the price level gap. While the model maintains high explanatory power (because

of the high number of variables), the inclusion of these variables introduces several issues.

Table 6.5: Estimation Results for the Extended Price Equation (with GAP)

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
<i>Dependent variable: Price growth (Δp_t)</i>					
Price growth (Δp_t)		0.83	0.48	0.21	1 to 4
Wage growth (Δw_t)		0.17	0.15	0.00	0 to 4
Energy price (ΔRPE_t)		0.22	0.01	0.06	0 to 4
Energy $\times$ Gap interaction		-0.003	0.06	0.11	1 to 5
Food price (ΔRPF_t)		-0.24	0.39	0.10	0 to 4
Food $\times$ Gap interaction		-0.001	0.16	0.30	1 to 5
Supply shortages ($SHORT_t$)		0.17	0.01	0.06	0 to 4
Productivity trend ($\Delta gpty_t$)		0.003	0.97	0.97	1
Structural trend ($GAPMA_t$)		0.03	0.35	0.35	1
Adjusted R^2		0.934	<i>Restriction: $\Sigma \Delta p + \Sigma \Delta w = 1$</i>		

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted models. The sums of coefficients (Σ) are estimated from the restricted model satisfying the theoretical condition of nominal neutrality.

First, the sum of coefficients for food price shocks is negative (-0.24), which is counter-intuitive. This happens because energy and food prices moved almost identically during the crisis (see figure 4.2). Since these variables are highly correlated, the model cannot separate their individual effects. It assigns most of the weight to energy shocks and forces the food coefficient into negative values to balance the equation.

Second, the high number of variables and lags reduces the degrees of freedom, which makes the interaction terms jointly insignificant. To find the op-

timal lag length and solve this over-parameterization, we tested models with 1 to 4 lags using AIC and BIC on the same sample period.

The lag selection shows a trade-off: BIC suggests 1 lag, while AIC prefers 4 lags. The AIC has a local minimum at 2 lags. From an economic perspective, a 1-lag structure is insufficient to capture the delayed transmission of shocks in quarterly data. Therefore, we use the 2-lag version as a compromise. This choice stabilizes the coefficients and provides a more realistic dynamic structure. The results of this refined 2-lag specification are presented in Table 6.6. This modification improves the economic consistency of the results, as the food price coefficient becomes appropriately positive (0.04). Even with this more stable parameterization and increased degrees of freedom, the interaction terms between commodity shocks and the price level gap remain statistically insignificant. This confirms that the lack of significance is a robust result and not an artifact of the chosen lag structure.

Table 6.6: Estimation Results for the Extended Price Equation (2 Lags, with GAP)

Variable		Sum (Σ)	p -stat (sum)	p -stat (joint)	Lags
<i>Dependent variable: Price growth (Δp_t)</i>					
Price	growth	0.17	0.61	0.43	1 to 2
(Δp_t)					
Wage	growth	0.83	0.00	0.00	0 to 2
(Δw_t)					
Energy	price	0.45	0.00	0.00	0 to 2
(ΔRPE_t)					
Energy $\times$ Gap in-		-0.00	0.13	0.36	1 to 3
teraction					
Food	price	0.04	0.67	0.17	0 to 2
(ΔRPF_t)					
Food $\times$ Gap inter-		0.00	0.25	0.23	1 to 3
action					
Supply	shortages	0.27	0.00	0.00	0 to 2
($SHORT_t$)					
Productivity		-0.02	0.31	0.31	1
trend ($\Delta gpty_t$)					
Structural	trend	-0.01	0.44	0.44	1
($GAPMA_t$)					
Adjusted R^2		0.893	<i>Restriction: $\Sigma \Delta p + \Sigma \Delta w = 1$</i>		

Notes: The p -stat (sum) and p -stat (joint) values are derived from the unrestricted models. The sums of coefficients (Σ) are estimated from the restricted model satisfying the theoretical condition of nominal neutrality. Lag structure reduced to 2 periods.

6.3.2 Discussion on effect of price level

While the extended model does not statistically identify the interaction effects, the baseline coefficients remain consistent with a higher sensitivity of inflation to energy shocks in the Czech Republic.

The estimated energy coefficient (Table 6.2) is relatively large in a European context. This difference is likely related to structural characteristics of the economy.

First, the Czech economy exhibits higher energy intensity due to its relatively large industrial sector. Second, essential items such as energy and food represent a larger share of the CPI basket in lower-price-level economies. As

a result, similar increases in global commodity prices translate into stronger domestic inflationary pressures.

In addition, the price level effect may be indirectly reflected in the persistence captured by the initial conditions. Price convergence, associated with the BS effect, is a gradual process, which may contribute to sustained inflation even after the dissipation of external shocks.

Residual diagnostics also do not suggest the presence of systematic non-linear patterns related to commodity shocks. Overall, while the direct effect of the price level gap is not statistically identified, the results remain consistent with broader structural differences in inflation transmission.

Chapter 7

Discussion and Policy Implications

The empirical results indicate that the post-pandemic inflation episode in the Czech Republic was primarily driven by external cost-push shocks, rather than by a self-reinforcing wage–price spiral. Despite a historically tight labor market, the estimated catch-up coefficient is negative, suggesting that past inflation surprises did not translate into higher nominal wage growth.

This finding places the Czech inflation experience within the broader European context. The dominance of supply-side energy costs and the absence of a strong wage-price spiral align closely with findings for Germany (Menz 2024), Italy (Pisani & Tagliabracci 2023), and France (Aldama *et al.* 2024). Consequently, the Czech dynamics stand in contrast to the US and Canadian experiences, where labor market tightness and rising nominal wages acted as drivers in sustaining inflation (Blanchard & Bernanke 2023; Bounajm *et al.* 2024).

The fact that wage dynamics did not significantly amplify inflationary pressures in the Czech Republic indicates that a substantial portion of the purchasing power loss was absorbed directly by households. As discussed in the empirical section, this counterintuitive result can be partially attributed to the specifics of the Czech labor market. The absence of automatic wage indexation and the highly decentralized nature of union bargaining mean that wage growth relies heavily on individual firm performance. During the cost shock, mainly in a Small Open Economy (SOE), firms faced financial constraints and therefore froze base wages and cut bonuses (Babecký *et al.* 2017). Furthermore, during that period, the CNB reported that nominal wage growth was moderate, leading to a large drop in real wages (Czech National Bank 2024b).

The Czech environment differs from economies with rigid wage-setting frame-

works. For example, De Walque & Lejeune (2024) shows that Belgium's automatic wage indexation led to significantly higher persistence of inflation. In the domestic context, the decentralized bargaining process weakened this transmission mechanism, preventing external shocks from fully passing through to labor costs.

In addition to institutional factors, behavioral responses could also play a role. The macroeconomic uncertainty that stemmed from the energy crisis and the geopolitical situation likely shifted employee preferences from wage maximization to job security. To avoid losing their jobs during a potential recession, workers accepted lower living standards instead of pushing for high nominal wage growth (Czech National Bank 2024a).

Furthermore, the high baseline pass-through of energy shocks estimated in our model aligns with the hypotheses raised in the literature regarding structural vulnerabilities. As noted by Coutinho & Licchetta (2023) and Bigerna (2023), industrial-oriented economies suffer disproportionately during energy crises. The high energy intensity of the Czech Republic served as a structural amplifier for global commodity spikes, explaining why the initial inflation surge was much steeper than in service-oriented economies.

From a policy perspective, these findings imply that inflation can reach and remain at high levels even in the absence of strong second-round wage effects. Consequently, monetary policy in such an environment should not rely solely on indicators of labor market tightness as signals of inflationary pressure. Greater emphasis should be placed on the transmission of external cost shocks and the structural composition of the economy.

Since wage-setting behavior did not act as the primary amplifier, aggressive monetary tightening aimed at cooling aggregate demand may involve higher output costs than in demand-driven economies like the US. In this context, targeted fiscal policies might be a more effective initial defense against pure commodity price shocks. As demonstrated by Ghomi *et al.* (2024), government interventions such as energy price caps successfully decoupled parts of the domestic price index from international volatility in Spain. For catching-up, highly industrialized economies, combining standard monetary policy with targeted fiscal or regulatory policies that directly address the underlying sources of the shock is therefore a useful approach when managing severe exogenous supply shocks in a way that limits adverse effects on the labor market.

Chapter 8

Conclusion

This thesis applied the BB framework to Czech data to identify and decompose the main drivers of the post-pandemic inflation episode. The empirical results highlight three main findings. First, external commodity shocks, particularly energy prices, played a significant role in triggering the inflation surge, exhibiting a notably higher pass-through than in other European economies. Second, inflation persistence remains substantial, indicating that domestic price-setting behavior is largely backward-looking. Third, while labor market tightness constitutes an important driver of wage dynamics, its overall contribution to the inflationary peak was secondary to external factors.

Importantly, the estimated catch-up coefficient is negative, in contrast to other country applications of the BB framework, where this coefficient is typically positive. This implies that Czech employees did not succeed in obtaining compensation for realized unexpected inflation, meaning past inflation surprises did not translate into higher nominal wage growth. This dynamic likely reflects the cost-push nature of the shock, which simultaneously weakened firms' financial conditions and limited their capacity to raise wages. Furthermore, this negative coefficient might capture the specific timing and rigidity of wage contracts, or the effect of omitted fiscal transfers (such as government energy subsidies) that partially compensated households outside the standard wage-bargaining process.

In addition to the baseline analysis, this thesis included an exploratory extension to test whether the domestic price level gap acts as an amplification mechanism. No statistically significant evidence of state-dependent effects was found. This result is robust across alternative specifications and reflects both data limitations and identification challenges, as the price level gap captures

multiple correlated structural features. Therefore, the role of the price level as an independent amplification mechanism cannot be conclusively established within the current empirical framework. From a methodological perspective, these results emphasize the importance of careful interpretation when extending semi-structural models with additional state variables.

Ultimately, the Czech experience demonstrates that severe cost-push inflation can persist and reach extreme levels without being fueled by a traditional wage–price spiral. Future research could build on this analysis by employing cross-country panel data or alternative identification strategies to better isolate the role of price-level differences in inflation dynamics.

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Appendix A

Data Comparison

This section provides a detailed comparison of the data series used in this thesis relative to the original US model. Table A.1 summarizes the definitions and sources for the endogenous variables, while Table A.2 details the exogenous variables.

Table A.1: Comparison of Variables: Endogenous Variables

Variable	Name	Original BB Model (US)	Czech Adaptation (CZ)
<i>gcpi</i>	Price inflation	Quarterly annualized rates of change in the Consumer Price Index (CPI), Bureau of Labor Statistics (BLS).	Seasonally adjusted quarterly annualized rates of change in the national Consumer Price Index (CPI), Czech Statistical Office (CZSO).
<i>gw</i>	Wage inflation	Rate of growth of nominal wages, measured by the change in the Employment Cost Index (ECI), Bureau of Labor Statistics (BLS).	Quarterly annualized rates of change in compensation per employee, European Central Bank (ECB).
<i>cf1</i>	Short-term inflation expectation	1-year-ahead inflation expectations constructed by the Federal Reserve Bank of Cleveland.	1-year-ahead inflation expectations of non-financial firms' managers, Czech National Bank (CNB).
<i>cf10 / cf3</i>	Long-term inflation expectation	10-year inflation expectations constructed by the Federal Reserve Bank of Cleveland.	3-year-ahead inflation expectations of financial market analysts, Czech National Bank (CNB).
—	Catch-up term	Losses to workers' purchasing power, measured by the 4-quarter average of CPI inflation minus the 1-year expectation 4 quarters earlier.	Realized inflation surprise measured as the gap between past inflation and past expectations.

Table A.2: Comparison of Variables: Exogenous Variables

Variable	Name	Original BB Model (US)	Czech Adaptation (CZ)
<i>grpe_cpi</i>	Relative energy price	Rate of change of the ratio of CPI energy prices to the ECI.	Quarterly annualized growth of relative energy prices (ratio of CPI energy to compensation per employee), Czech Statistical Office (CZSO).
<i>grpfcpi</i>	Relative food price	Rate of change of the ratio of CPI food prices to the ECI.	Quarterly annualized growth of relative food prices (ratio of CPI food to compensation per employee), Czech Statistical Office (CZSO).
<i>vu</i>	Labor market tightness	Ratio of job vacancies to unemployment, Bureau of Labor Statistics (BLS).	Ratio of job vacancies to unplaced job seekers, Ministry of Labor and Social Affairs (MoLSA) and Czech National Bank (CNB).
—	Shortage indicator	Index of shortages based on Google searches for the term "shortage".	Global Supply Chain Pressure Index (GSCPI), standardized, Federal Reserve Bank of New York (NYFED).
<i>gpty</i>	Productivity	8-quarter moving average of log changes of nonfarm business value added divided by nonfarm employee hours, Bureau of Labor Statistics (BLS).	8-quarter moving average of quarterly annualized growth in labor productivity per person, European Central Bank (ECB).

Appendix B

Diagnostics

This appendix presents diagnostic tests for the empirical model. Table B.1 reports ADF unit root tests to verify time series stationarity. Table B.2 summarizes the Wald test results regarding the theoretical homogeneity constraint.

Table B.1: Unit Root Diagnostics: Augmented Dickey-Fuller (ADF) Test Results

Variable	Description	Sample Period	p-value
<i>Endogenous Variables</i>			
gw	Wage growth (nominal)	1995Q2 – 2025Q3	0.001
gcpi	CPI inflation (annualized)	1996Q2 – 2025Q4	0.002
ghicp	HICP inflation (annualized)	1996Q2 – 2025Q4	0.002
cf1	Short-term expectations (1Y)	1999Q4 – 2025Q4	0.003
cf3	Long-term expectations (3Y)	1999Q4 – 2025Q4	0.019
diff_cpi	Catch-up term (CPI)	2000Q4 – 2025Q4	0.004
diff_hicp	Catch-up term (HICP)	2000Q4 – 2025Q4	0.006
<i>Exogenous and Supply Shock Variables</i>			
grpe_cpi	Rel. energy price growth (CPI)	2000Q1 – 2025Q3	0.010
grp_f_cpi	Rel. food price growth (CPI)	2000Q1 – 2025Q3	0.059
grpe_hicp	Rel. energy price growth (HICP)	2000Q1 – 2025Q3	0.026
grp_f_hicp	Rel. food price growth (HICP)	2000Q1 – 2025Q3	0.218
vu	Labor market tightness (V/U)	1993Q4 – 2025Q4	0.273
u_gap	Unemployment gap	1993Q1 – 2025Q4	0.419
shortage	Supply shortages (GSCPI)	1998Q1 – 2025Q4	0.034
gpty	Productivity trend (8Q MA)	1997Q1 – 2025Q3	0.124
gap	Price level gap (GAP_t)	1995Q4 – 2025Q4	0.664
gap_ma8	Smoothed price level gap	1997Q3 – 2025Q4	0.562

Note: The null hypothesis of the ADF test is that the series has a unit root (non-stationary). p-values below 0.05 indicate stationarity at the 5% level.

Table B.2: Homogeneity Constraint Diagnostics: Wald Test p-values

Country	gw	gcpi	cf1	cf10/cf3
USA	0.0080	0.4592	0.3114	0.1742
Euro Area	0.1576	0.5152	0.0017	0.9524
United Kingdom	0.9186	0.6591	0.3280	0.1338
France	0.5318	0.1166	0.1380	0.6576
Germany	0.0074	0.0003	0.4776	0.7548
Spain	0.0014	0.0178	0.4817	0.2338
Italy	0.4355	0.4256	0.0054	0.7197
Netherlands	0.2456	0.1669	0.2581	0.3173
Belgium	0.7941	0.0001	0.0014	0.1882
Canada	0.4873	0.0008	0.0782	0.9524
Czechia	0.0043	0.0019	0.6629	0.0008

Note: The table reports p-values for the Wald test of the homogeneity restriction. A p-value > 0.05 indicates that the data do not reject the theoretical macroeconomic restriction at the 5% significance level. The underlying coefficients are estimated on the full sample. Data for other countries are from (Bernanke & Blanchard 2024).

Appendix C

Sensitivity and Robustness Analysis

C.1 Unrestricted Model Estimates (Main model)

This section reports estimates from the unrestricted specifications of the model, without imposing theoretical restrictions, to assess the sensitivity of the results.

Table C.1: Wage Equation (CPI): Unrestricted Specification

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw	0.44	0.09	0.01	-1 to -4
cf1	-0.64	0.21	0.02	-1 to -4
catch-up	-0.78	0.07	0.01	-1 to -4
v/u	3.18	0.01	0.00	-1 to -4
gpty	0.23	0.25	0.25	-1
R-squared	0.534			

Table C.2: Price Equation (CPI): Unrestricted Specification

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gcpi	0.34	0.18	0.00	1 to 4
gw	0.34	0.12	0.00	0 to 4
grpe	0.24	0.05	0.00	0 to 4
grpf	0.08	0.32	0.00	0 to 4
shortage	0.42	0.00	0.01	0 to 4
gpty	0.02	0.75	0.75	1
R-squared	0.900			

Table C.3: Short-term Inflation Expectations (CPI): Unrestricted Specification

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
cf1	0.84	0.00	0.00	1 to 4
cf3	0.12	0.27	0.35	0 to 4
gcpi	0.08	0.04	0.00	0 to 4
R-squared	0.977			

Table C.4: Long-term Inflation Expectations (CPI): Unrestricted Specification

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
cf3	0.89	0.00	0.00	1 to 4
gcpi	0.01	0.04	0.00	0 to 4
R-squared	0.919			

C.2 Robustness Checks: National CPI Data

In this section, we present the estimation results of the wage equation using alternative variables for labor market tightness and wage measures. Section C.2 focuses on national CPI data, while Section C.3 uses the European Harmonised Index of Consumer Prices (HICP).

C.2.1 Alternative Wage Measure: Compensation per Hour

Table C.5: Wage Equation (CPI): Compensation per Hour and V/U

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw (hourly)	0.06	0.88	0.00	-1 to -4
cf1	0.94	0.66	0.33	-1 to -4
v/u	3.35	0.06	0.02	-1 to -4
catch-up	-1.62	0.04	0.26	-1 to -4
gpty	0.17	0.72	0.72	-1
R-squared	0.474			

Note: The sum of coefficients are estimated from the Restricted Model, satisfying the theoretical condition $\Sigma gw + \Sigma cf1 = 1$. The p-stat (sum) and p-stat (joint) values are derived from the Unrestricted Model.

C.2.2 Alternative Labor Tightness Measure: Unemployment Rate Gap

In this robustness check, we replace the V/U ratio with the unemployment rate gap (u_gap), defined as the deviation of the unemployment rate from its 2019Q4 level.

Table C.6: Wage Equation (CPI): Compensation per Employee and U -gap

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw	0.06	0.94	0.01	-1 to -4
cfl	0.94	0.45	0.15	-1 to -4
u_gap	-0.78	0.00	0.00	-1 to -4
catch-up	-1.40	0.02	0.09	-1 to -4
gpty	0.52	0.01	0.01	-1
R-squared	0.609			

C.2.3 Combined Alternative Measures

Table C.7: Wage Equation (CPI): Compensation per Hour and U -gap

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw (hourly)	0.06	0.88	0.00	-1 to -4
cfl	0.94	0.51	0.27	-1 to -4
u_gap	-1.17	0.03	0.14	-1 to -4
catch-up	-1.74	0.03	0.25	-1 to -4
gpty	0.41	0.45	0.45	-1
R-squared	0.440			

C.3 Robustness Checks: European HICP Data

C.3.1 Baseline Specification: Compensation per Employee and V/U

Table C.8: Wage Equation (HICP): Baseline Measures (Employee and V/U)

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw	0.31	0.10	0.00	-1 to -4
cf1	0.69	0.24	0.10	-1 to -4
v/u	2.76	0.01	0.00	-1 to -4
catch-up	-1.19	0.20	0.07	-1 to -4
gpty	0.25	0.22	0.22	-1
R-squared	0.596			

C.3.2 Alternative Wage Measure: Compensation per Hour

This specification uses the growth of compensation per hour as the dependent variable while maintaining the baseline V/U ratio as the measure of labor market tightness, all applied to HICP data.

Table C.9: Wage Equation (HICP): Compensation per Hour and V/U

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw (hourly)	0.06	0.88	0.00	-1 to -4
cf1	0.94	0.62	0.57	-1 to -4
v/u	3.15	0.08	0.03	-1 to -4
catch-up	-1.32	0.09	0.38	-1 to -4
gpty	0.15	0.74	0.74	-1
R-squared	0.466			

C.3.3 Alternative Labor Tightness Measure: Unemployment Rate Gap

In this case, the V/U ratio is replaced by the unemployment rate gap, while keeping the compensation per employee as the wage measure under the HICP framework.

Table C.10: Wage Equation (HICP): Compensation per Employee and U -gap

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw	0.04	0.97	0.01	-1 to -4
cfl	0.96	0.42	0.34	-1 to -4
u_gap	-0.79	0.01	0.01	-1 to -4
catch-up	-1.22	0.05	0.22	-1 to -4
gpty	0.51	0.01	0.01	-1
R-squared	0.598			

C.3.4 Combined Alternative Measures

Finally, this specification combines both alternative measures: the growth of compensation per hour and the unemployment rate gap, using HICP as the price index.

Table C.11: Wage Equation (HICP): Compensation per Hour and U -gap

Variable	Sum of Coeff. (Σ)	p-stat (sum)	p-stat (joint)	Lags
gw (hourly)	0.06	0.86	0.00	-1 to -4
cfl	0.94	0.47	0.37	-1 to -4
u_gap	-1.13	0.03	0.16	-1 to -4
catch-up	-1.53	0.05	0.29	-1 to -4
gpty	0.35	0.51	0.51	-1
R-squared	0.436			

Wage Growth Comparison: Per Employee vs. Per Hour

(annualized quarter-on-quarter changes)

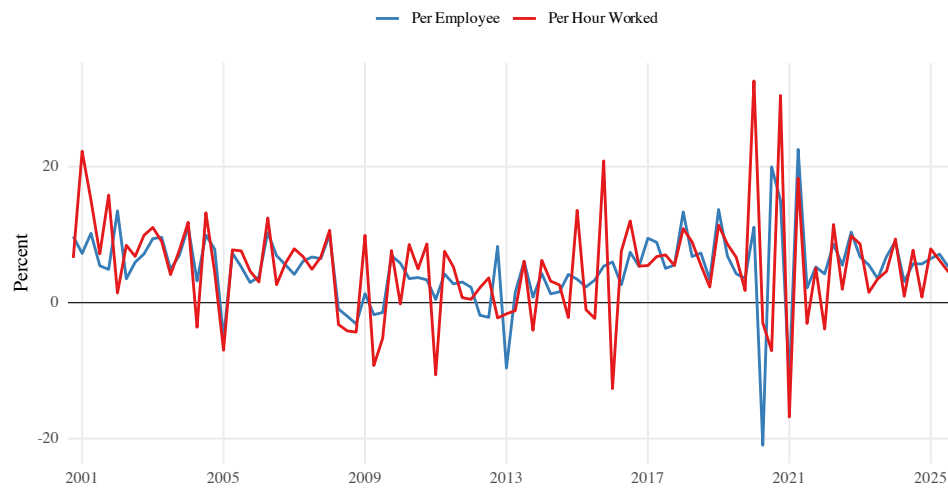


Figure C.1: Wage growth dynamics comparison.

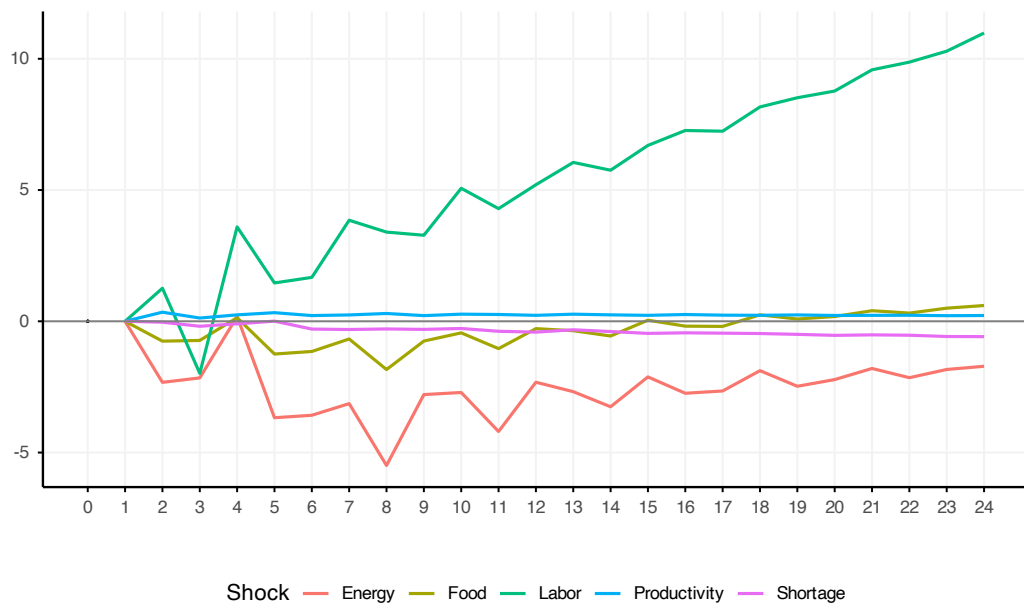
C.3.5 Alternative IRF's using Compensation per Hour

Figure C.2: Impulse Response Function of Wage growth

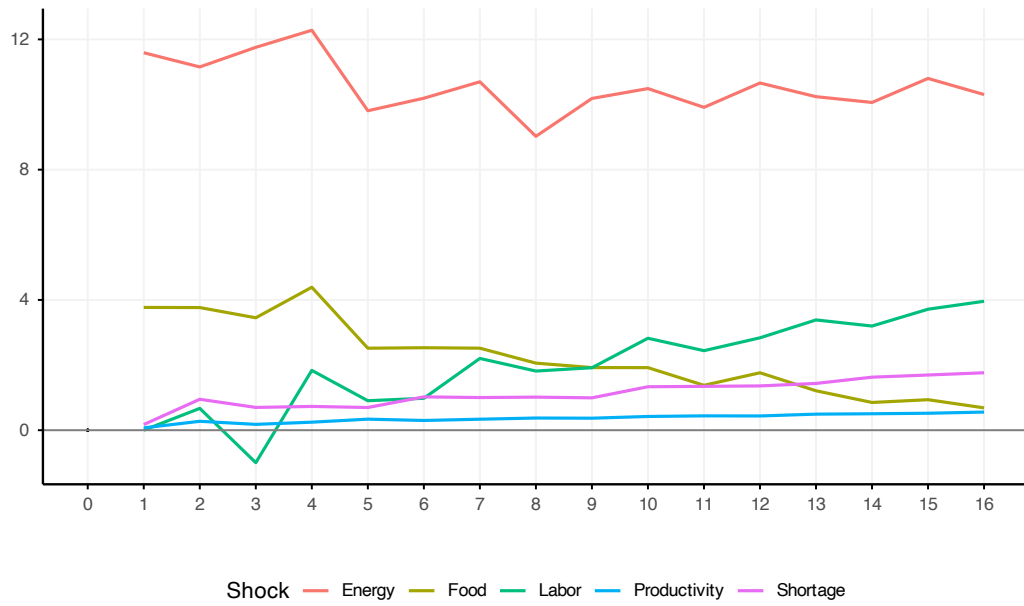


Figure C.3: Impulse Response Function of Inflation

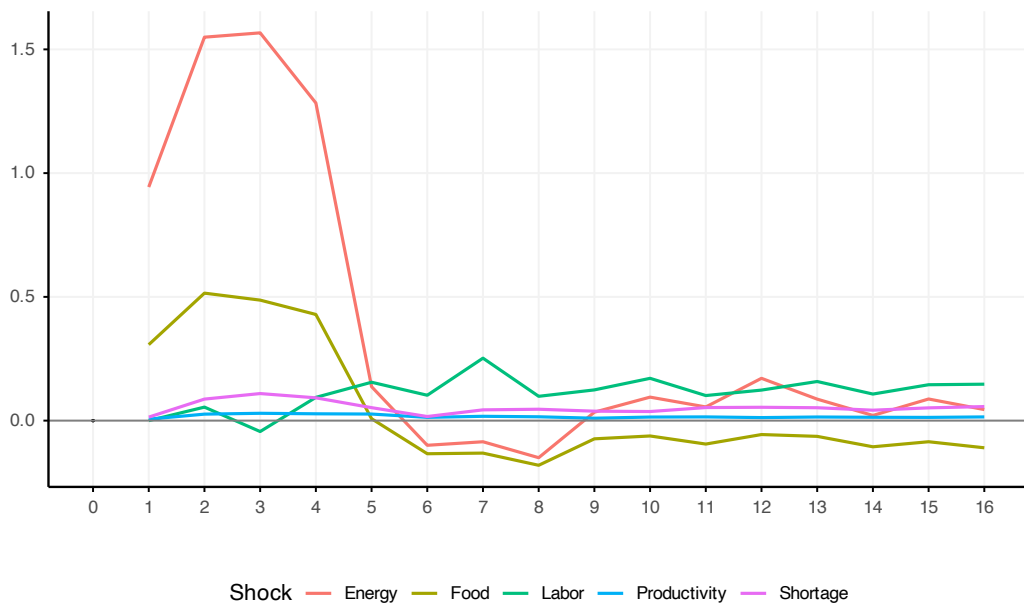


Figure C.4: IRF of Short-term inflation expectations

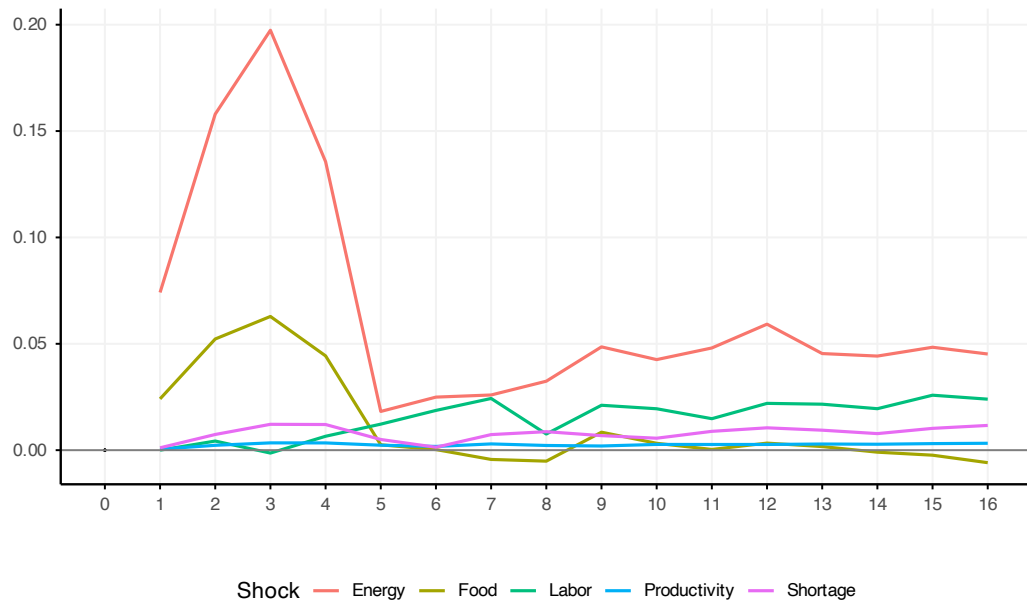


Figure C.5: IRF of Long-term inflation expectations

C.3.6 Alternative Decomposition using Compensation per Hour

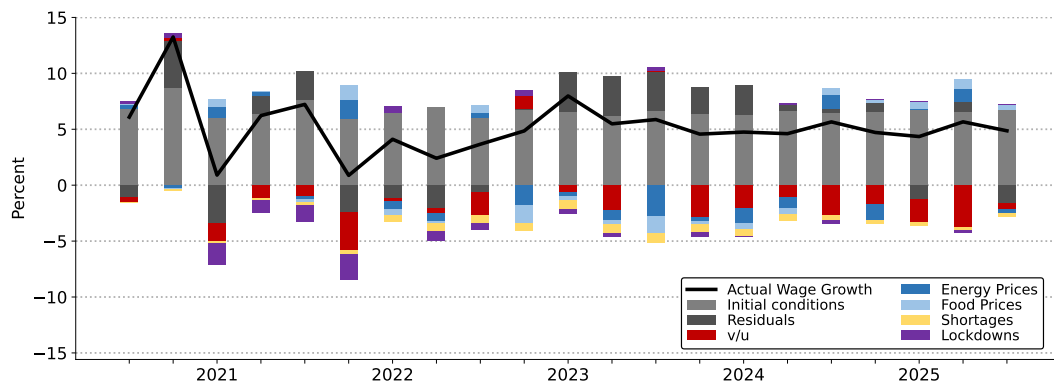


Figure C.6: Historical Decomposition of Wage growth

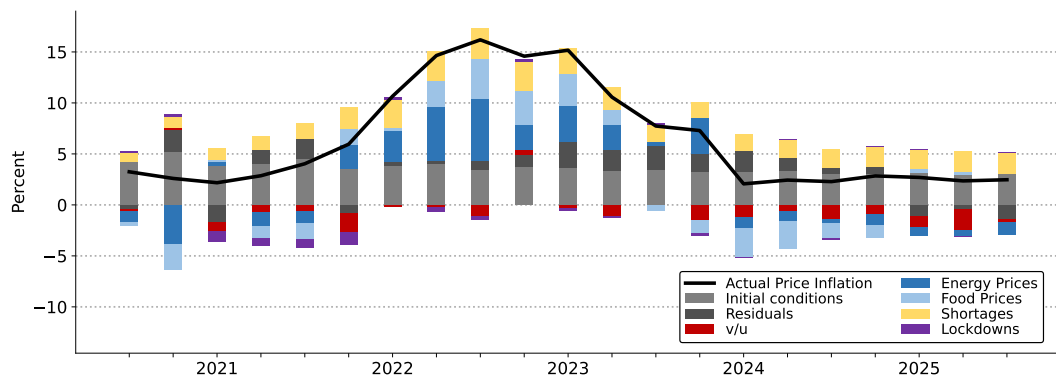


Figure C.7: Historical Decomposition of Inflation

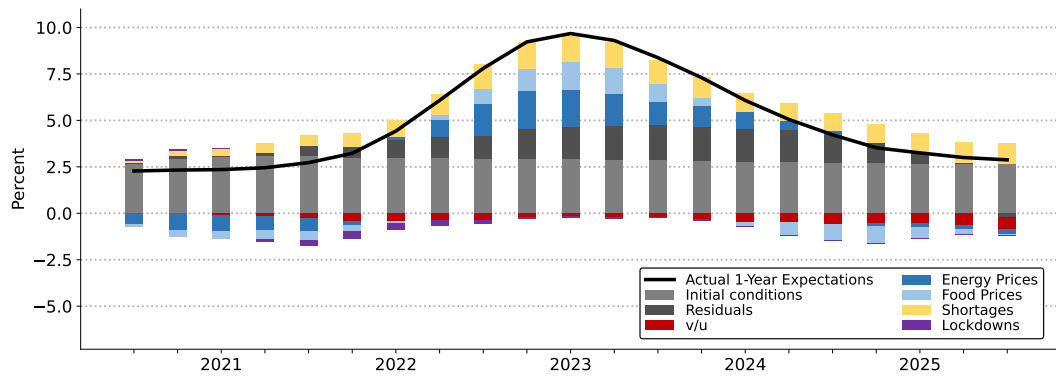


Figure C.8: Historical Decomposition of Short-term inflation expectations

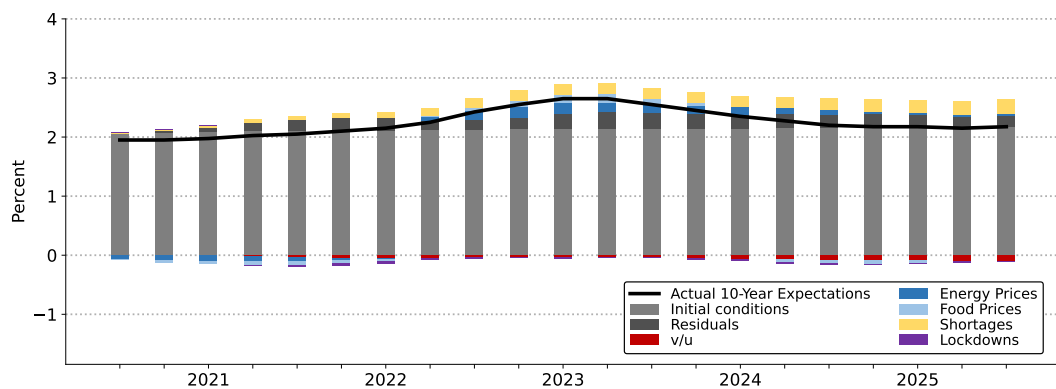


Figure C.9: Historical Decomposition of Long-term inflation expectations

Appendix D

Project's website

The source code and datasets used in this research are available in the following repository: <https://github.com/smirovaclav/thesis>